UNIVERSITAT JAUME I

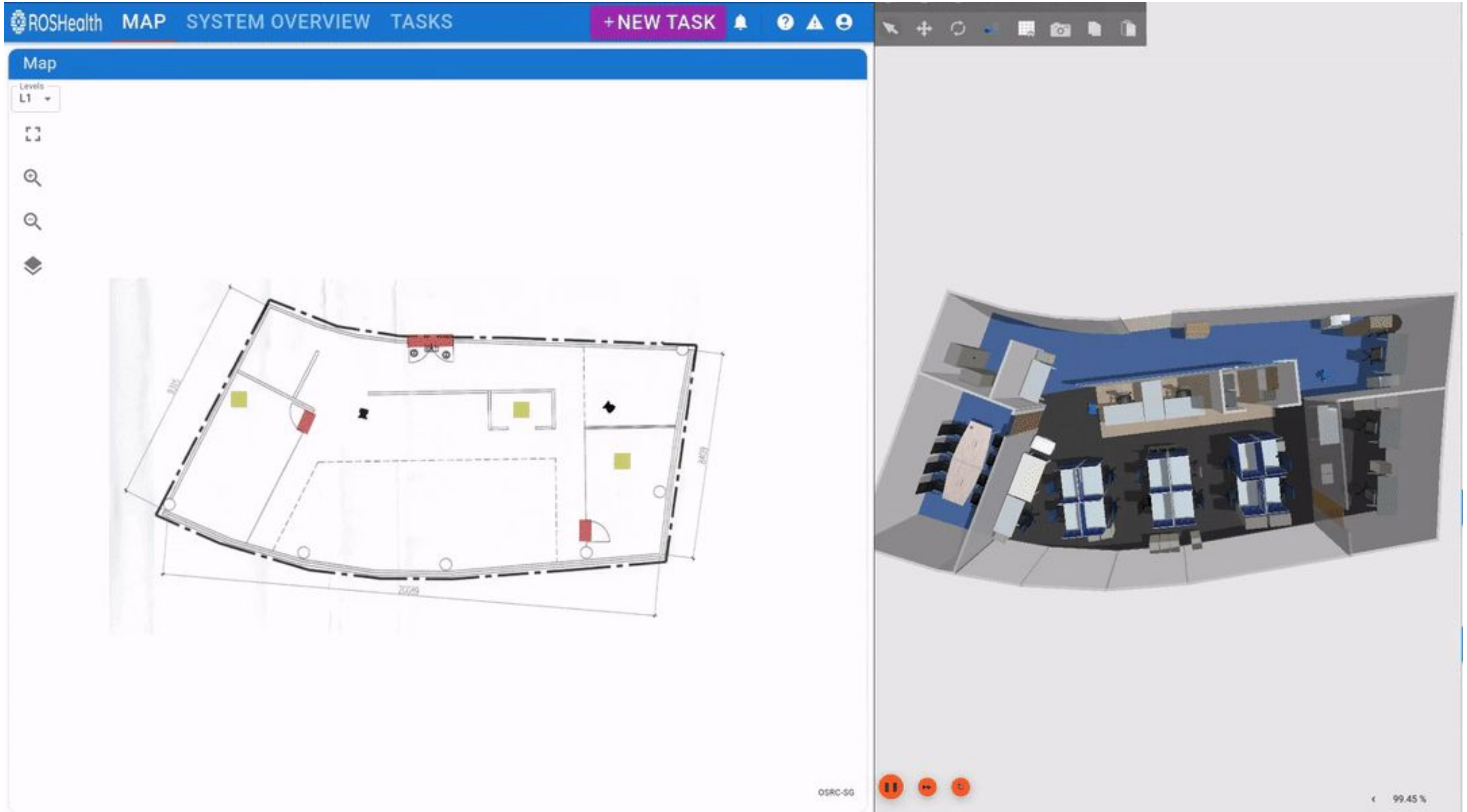
Intel·ligència Robòtica

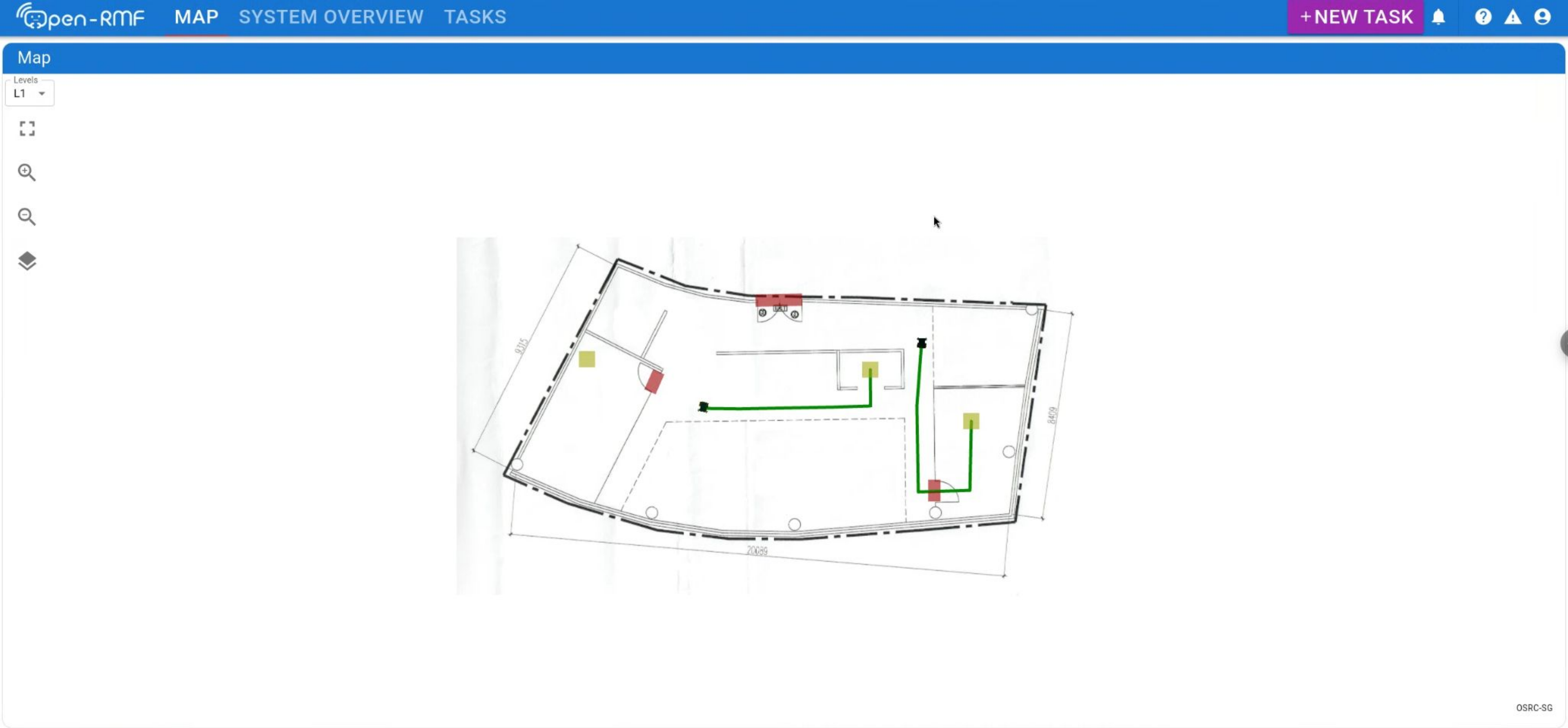
Open-RMF Web

Web-based interface to Open-RMF

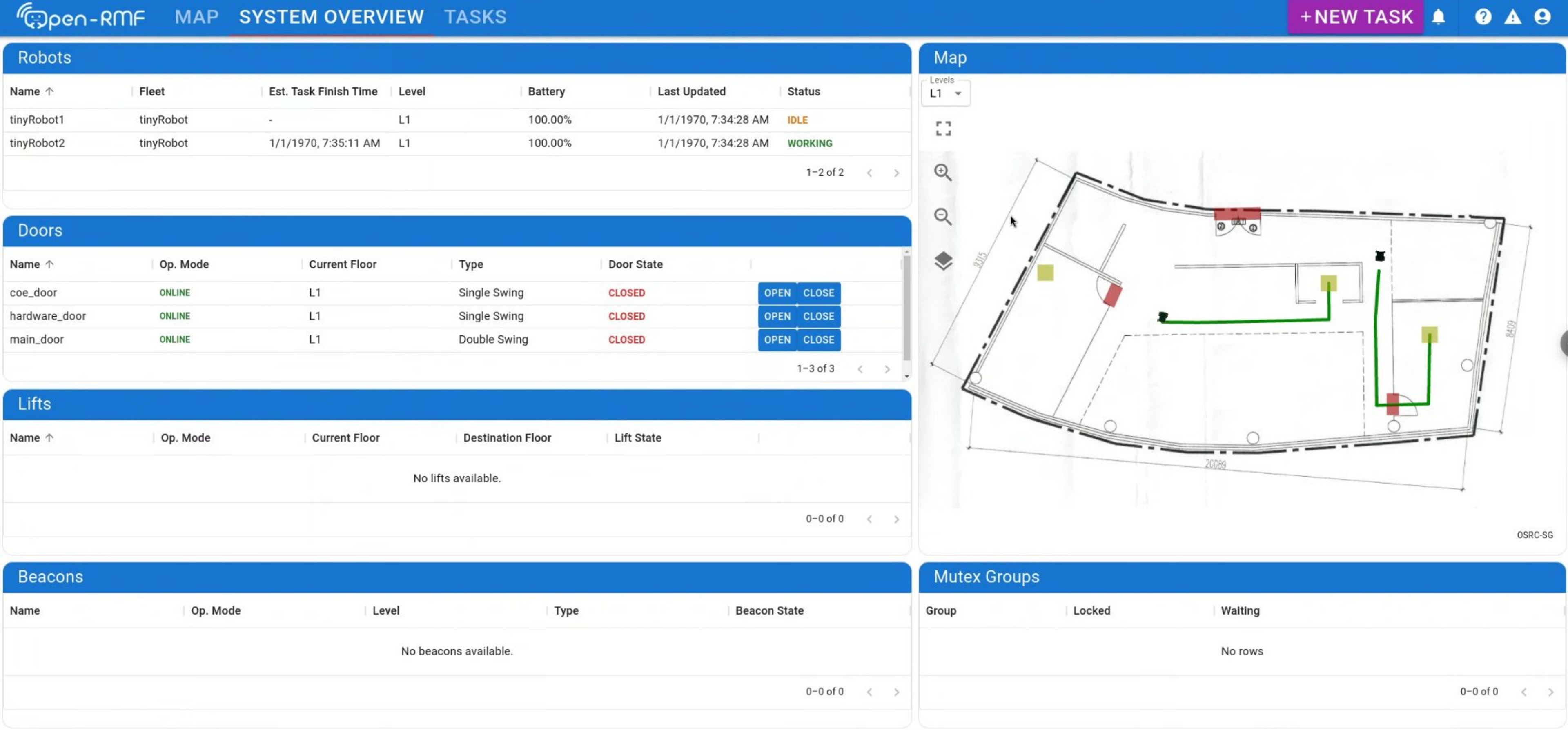
RMF Web

RMF Web is a collection of packages that provide a web-based interface for users to visualize and control all aspects of Open-RMF deployments.

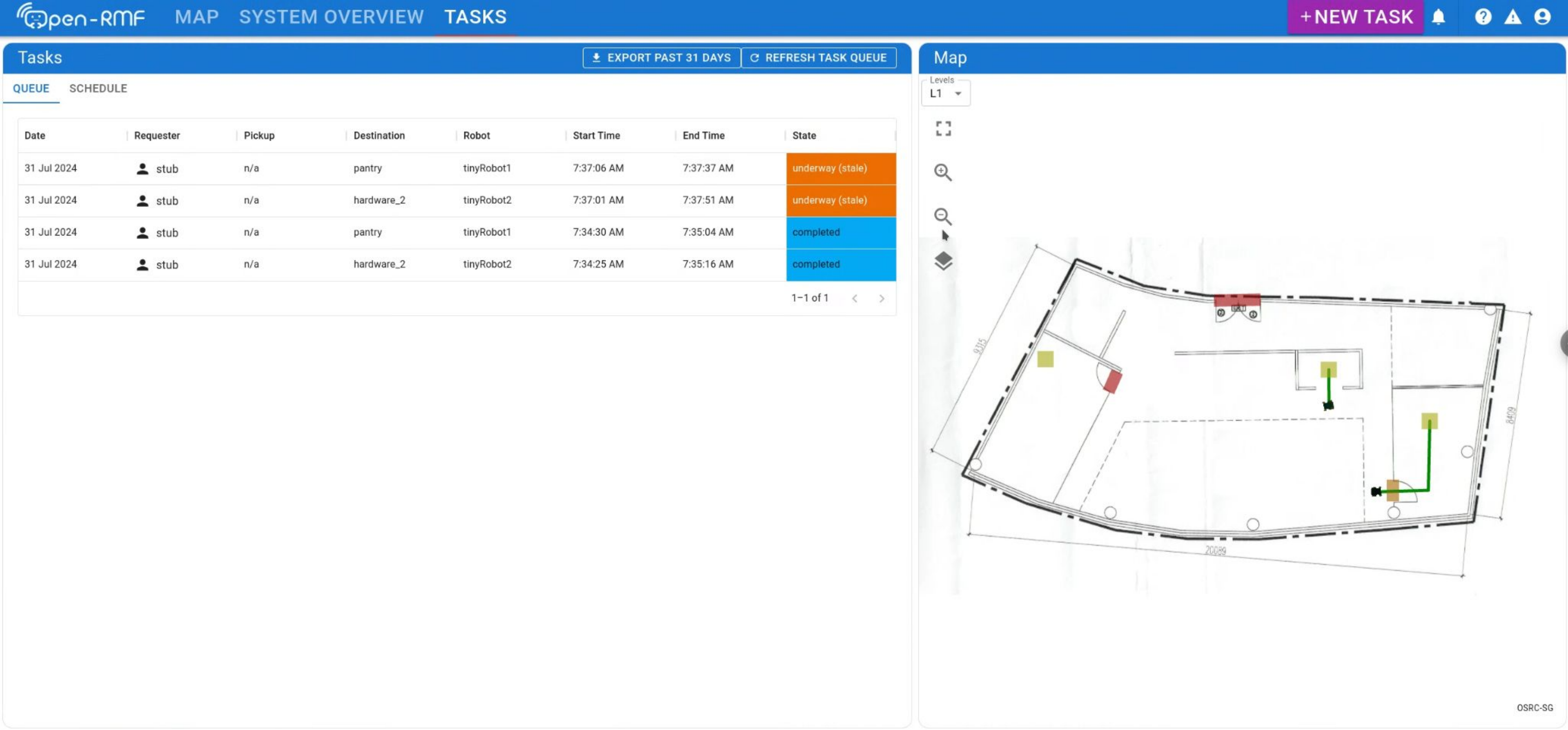




Open-RMF Web provides a comprehensive web application toolkit, with an API server, re-usable frontend components, as well as a customizable frontend dashboard.

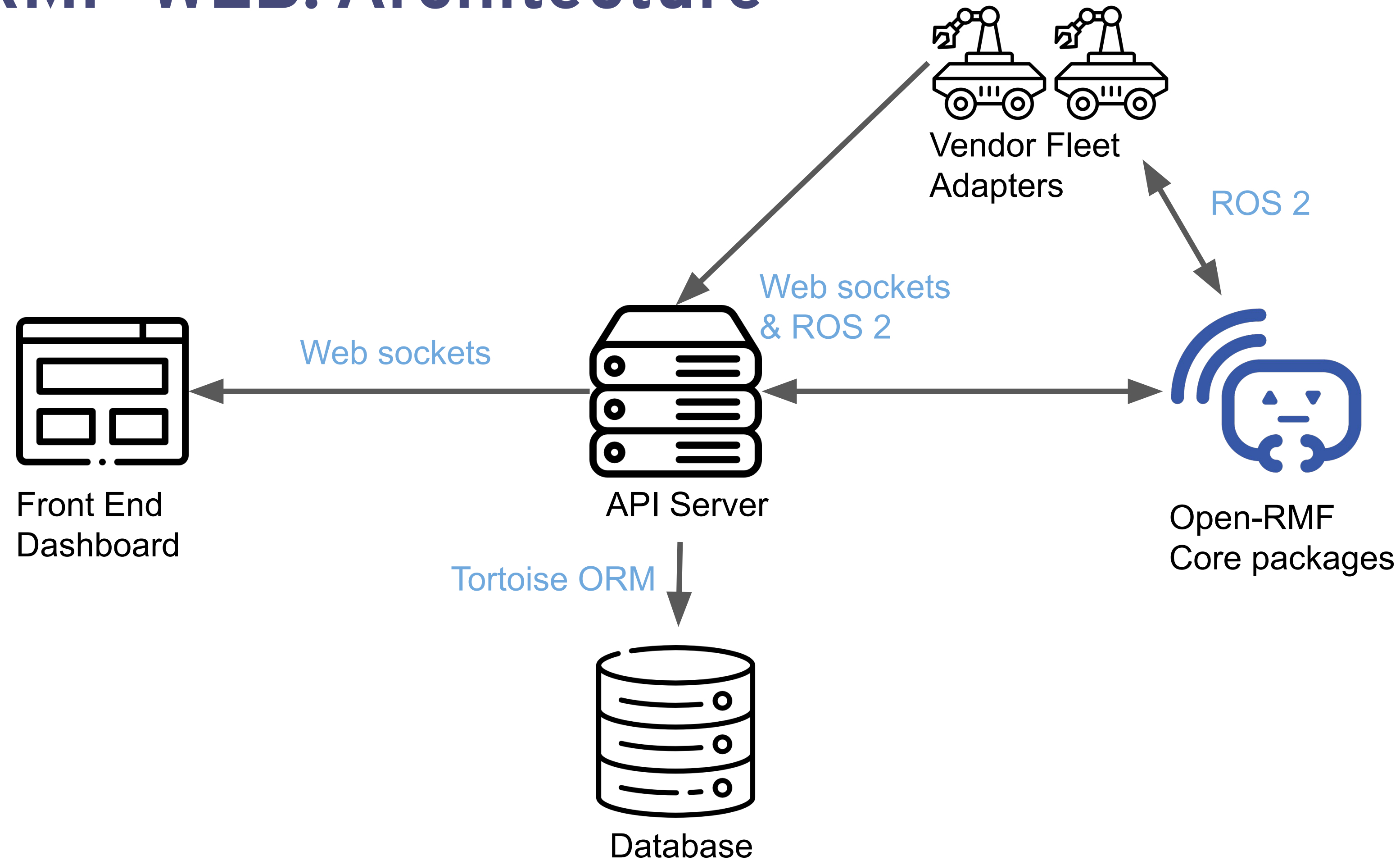


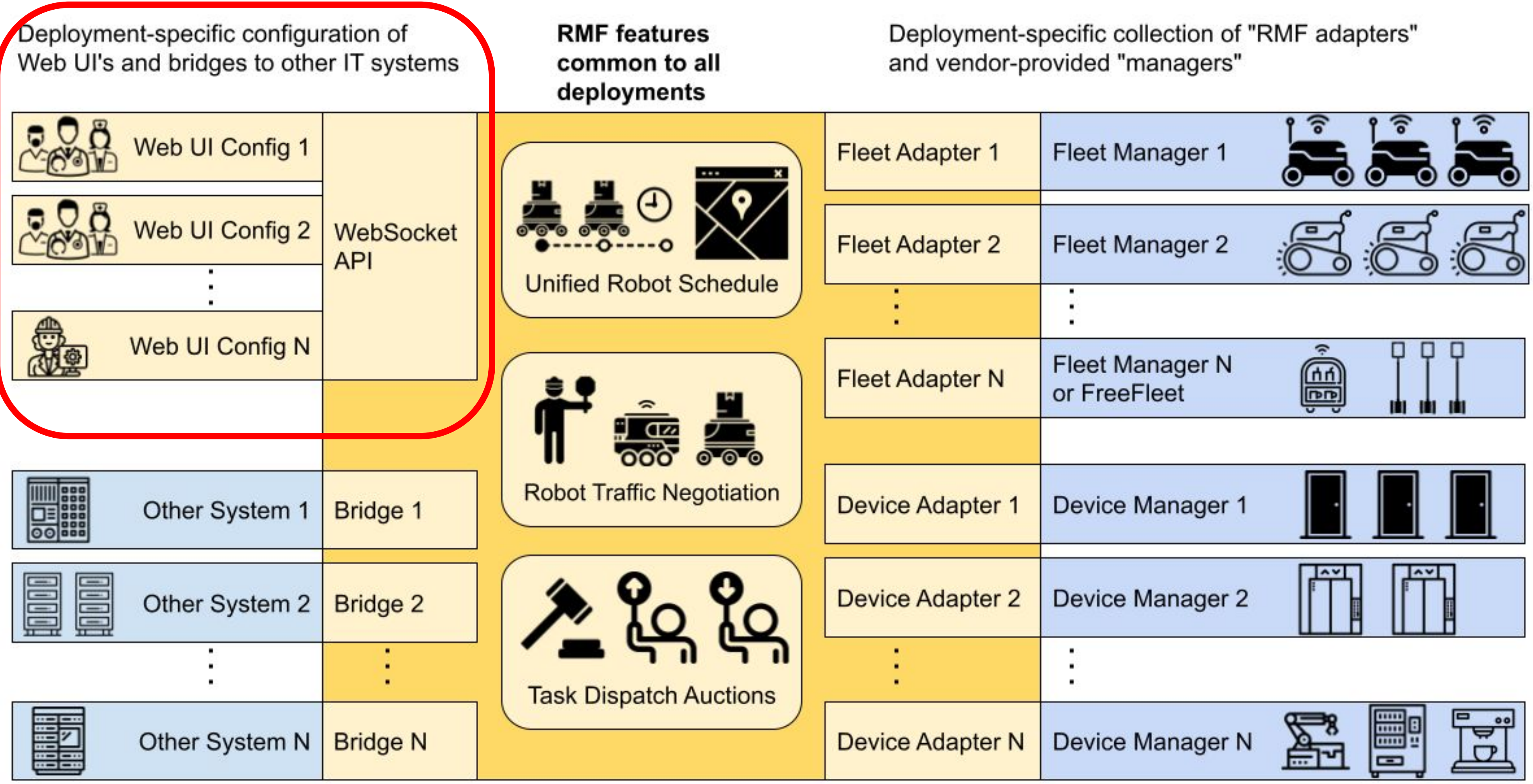
Open-RMF Web provides a comprehensive web application toolkit, with an API server, re-usable frontend components, as well as a customizable frontend dashboard.



Open-RMF Web provides a comprehensive web application toolkit, with an API server, re-usable frontend components, as well as a customizable frontend dashboard.

RMF-WEB: Architecture

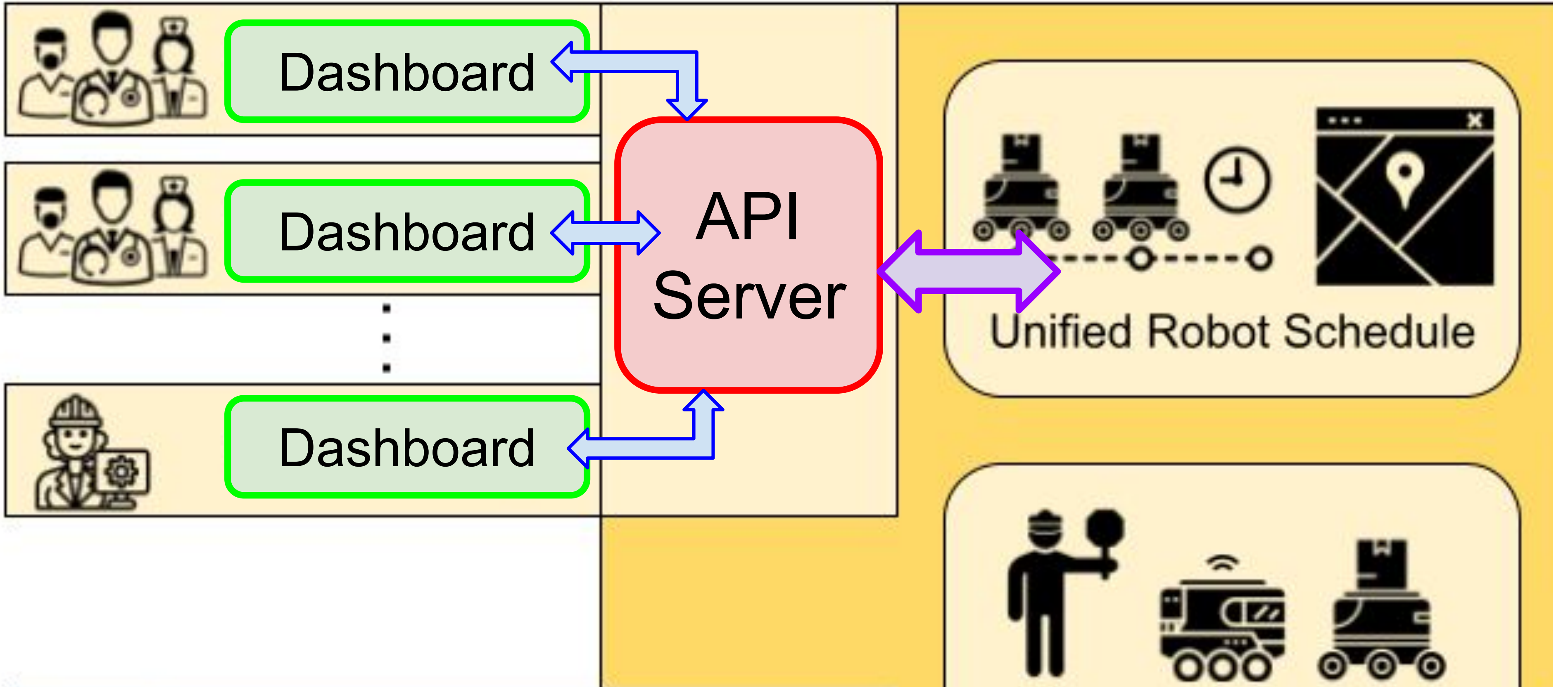




Deployment-specific configuration of Web UI's and bridges to other IT systems
RMF features common to all deployments
Deployment-specific collection of "RMF adapters" and vendor-provided "managers"
Web UI Config 1
Web UI Config 2
Web UI Config N
WebSocket API
Other System 1
Bridge 1
Other System 2
Bridge 2
Other System N
Bridge N
Unified Robot Schedule
Robot Traffic Negotiation
Task Dispatch Auctions
Fleet Adapter 1
Fleet Manager 1
Fleet Adapter 2
Fleet Manager 2
Fleet Adapter N
Fleet Manager N or FreeFleet
Device Adapter 1
Device Manager 1
Device Adapter 2
Device Manager 2
Device Adapter N
Device Manager N

Deployment-specific configuration of Web UI's and bridges to other IT systems

RMF features common to all deployments



Open-RMF API Server

This API server sets up the necessary endpoints with an Open-RMF deployment and enables the use of the web dashboard.

The server comes with the capability of logging to databases, as well as handling authentication and permissions.

Quick start with Docker

```
docker pull ghcr.io/open-rmf/rmf-web/api-server:jazzy-nightly

docker pull ghcr.io/open-rmf/rmf-web/demo-dashboard:jazzy-nightly
```

You need 3 terminals and the web browser:

Terminal 1: launch an Open-RMF demo

Terminal 2: launch the API server

Terminal 3: launch the dashboard

Web browser: the dashboard is accessible on localhost:3000 by default.

Terminal 1 - Open RMF demo

```
rocker --nvidia --x11 --name rmf_demos \
  -e ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST \
  --network host \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
   ros2 launch rmf_demos_gz office.launch.xml \
   server_uri:="ws://localhost:8000/_internal"
```

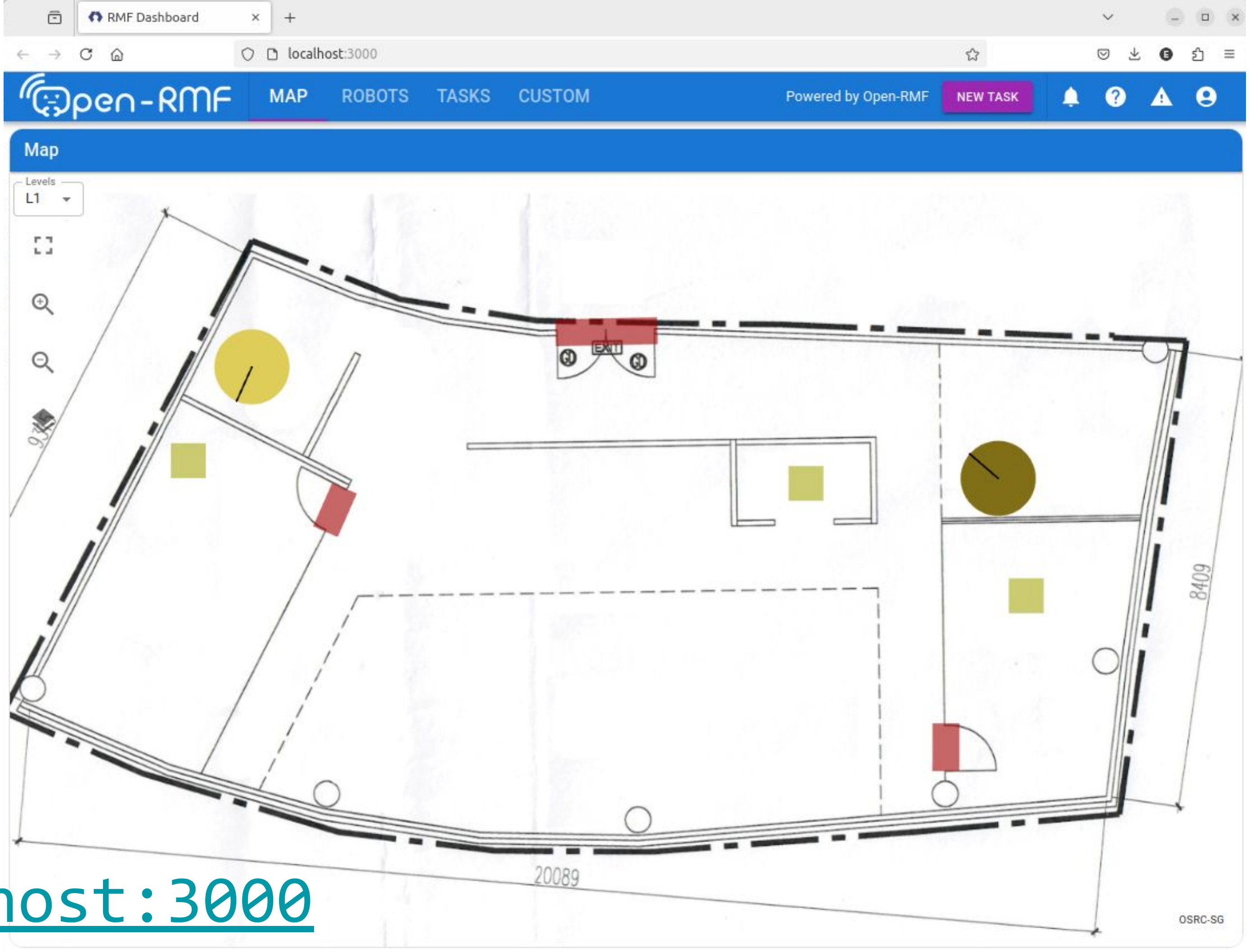
Terminal 2 - API Server

```
docker run --network host --rm -it \
  -e ROS_AUTOMATIC_DISCOVERY_RANGE=LOCALHOST \
  -e RMW_IMPLEMENTATION=rmw_cyclonedds_cpp \
   ghcr.io/open-rmf/rmf-web/api-server:jazzy-nightly
```

Terminal 3 - dashboard

```
docker run --network host --rm -it \
  -e RMF_SERVER_URL=http://localhost:8000 \
  -e TRAJECTORY_SERVER_URL=ws://localhost:8006 \
   ghcr.io/open-rmf/rmf-web/demo-dashboard:jazzy-nightly
```

Web browser



http://localhost:3000

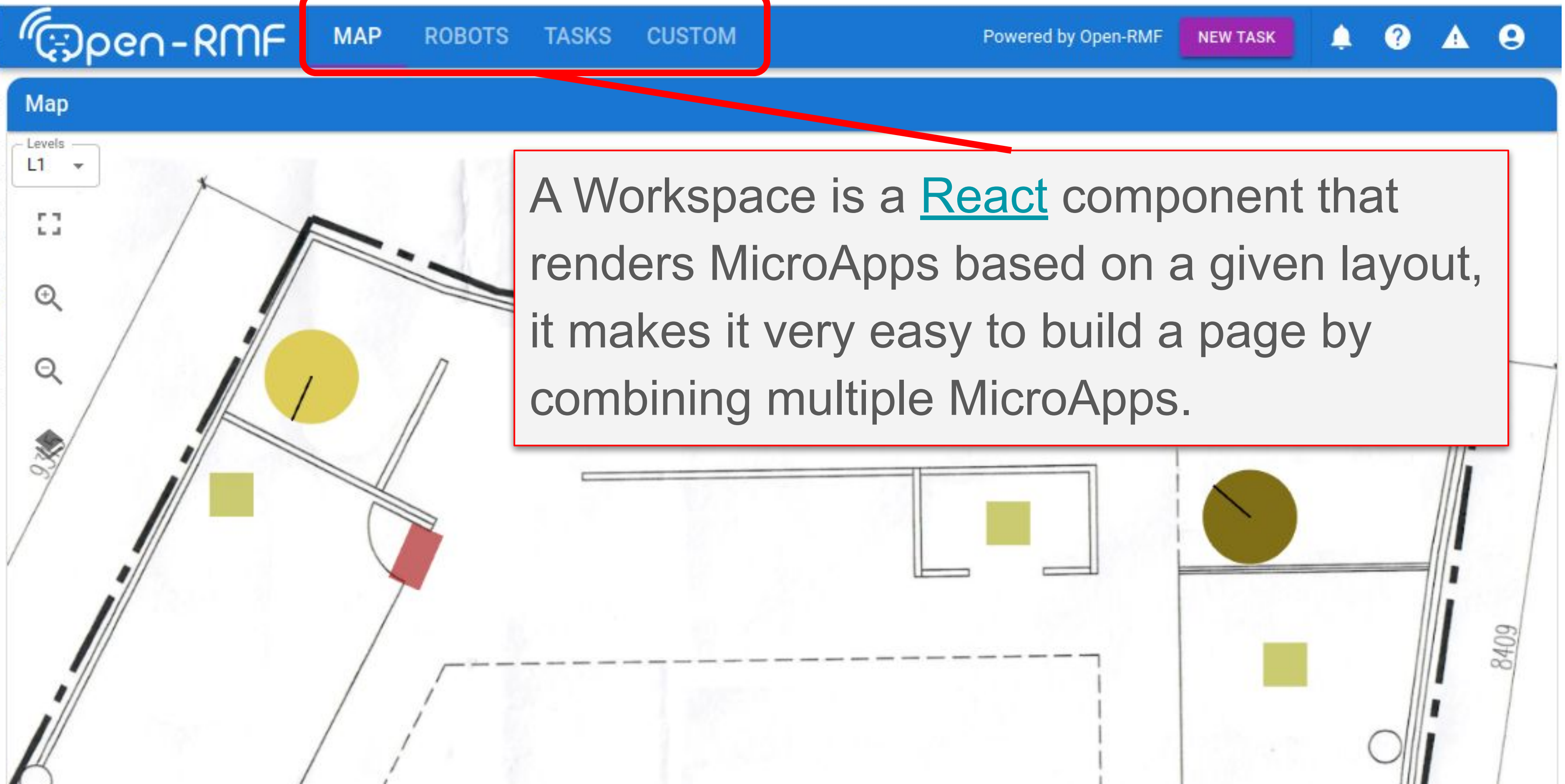
MicroApps

MicroApps are React components that can be used to build a page.
The dashboard comes with many different MicroApps, each serving a different purpose when interacting with an Open-RMF deployment.

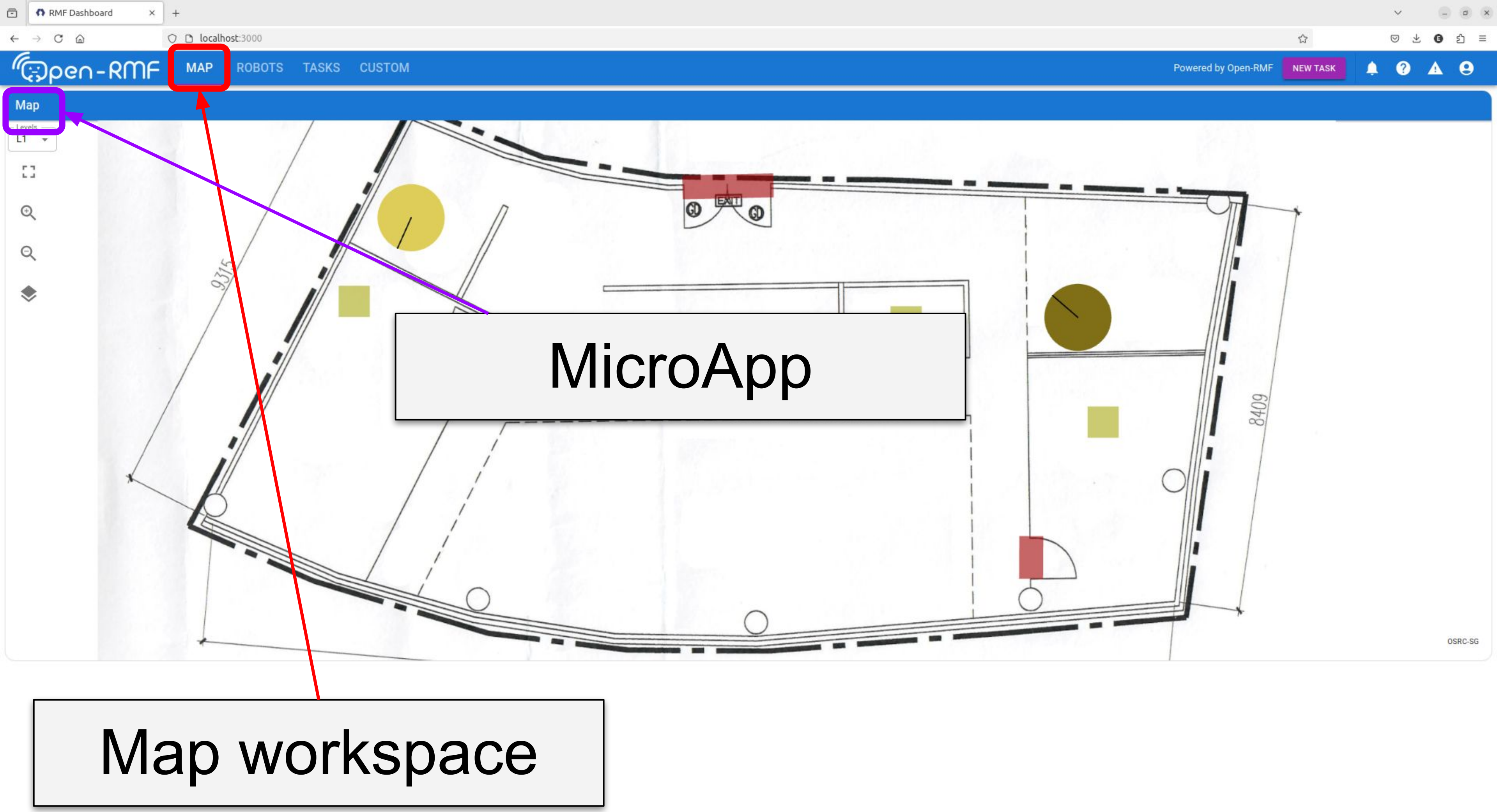
Some of the built in MicroApps are:

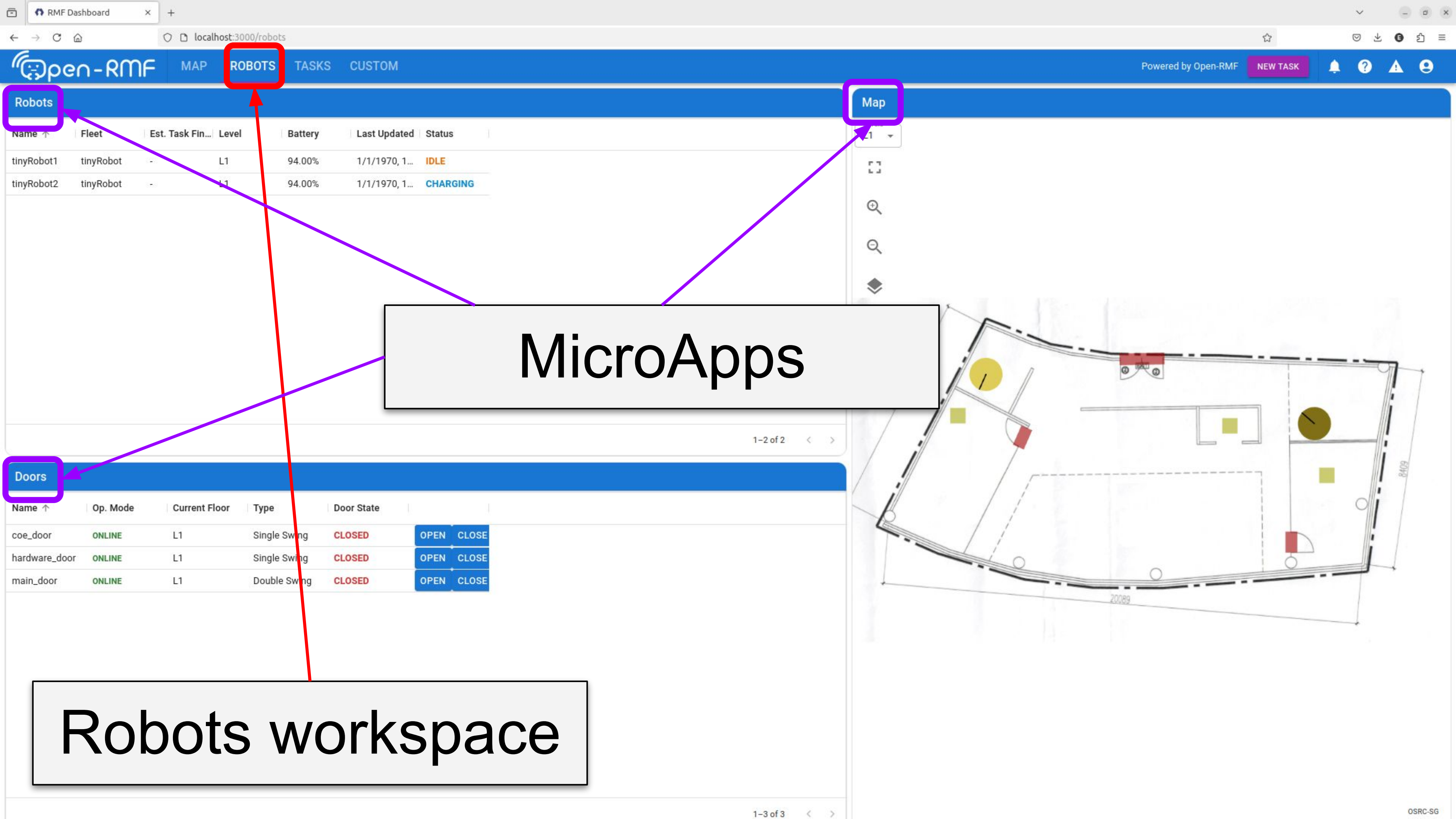
- Robots
- Map
- Doors
- Lifts
- Mutex Groups
- Tasks
- Beacons
- Robot Info
- Task Details
- Task Logs

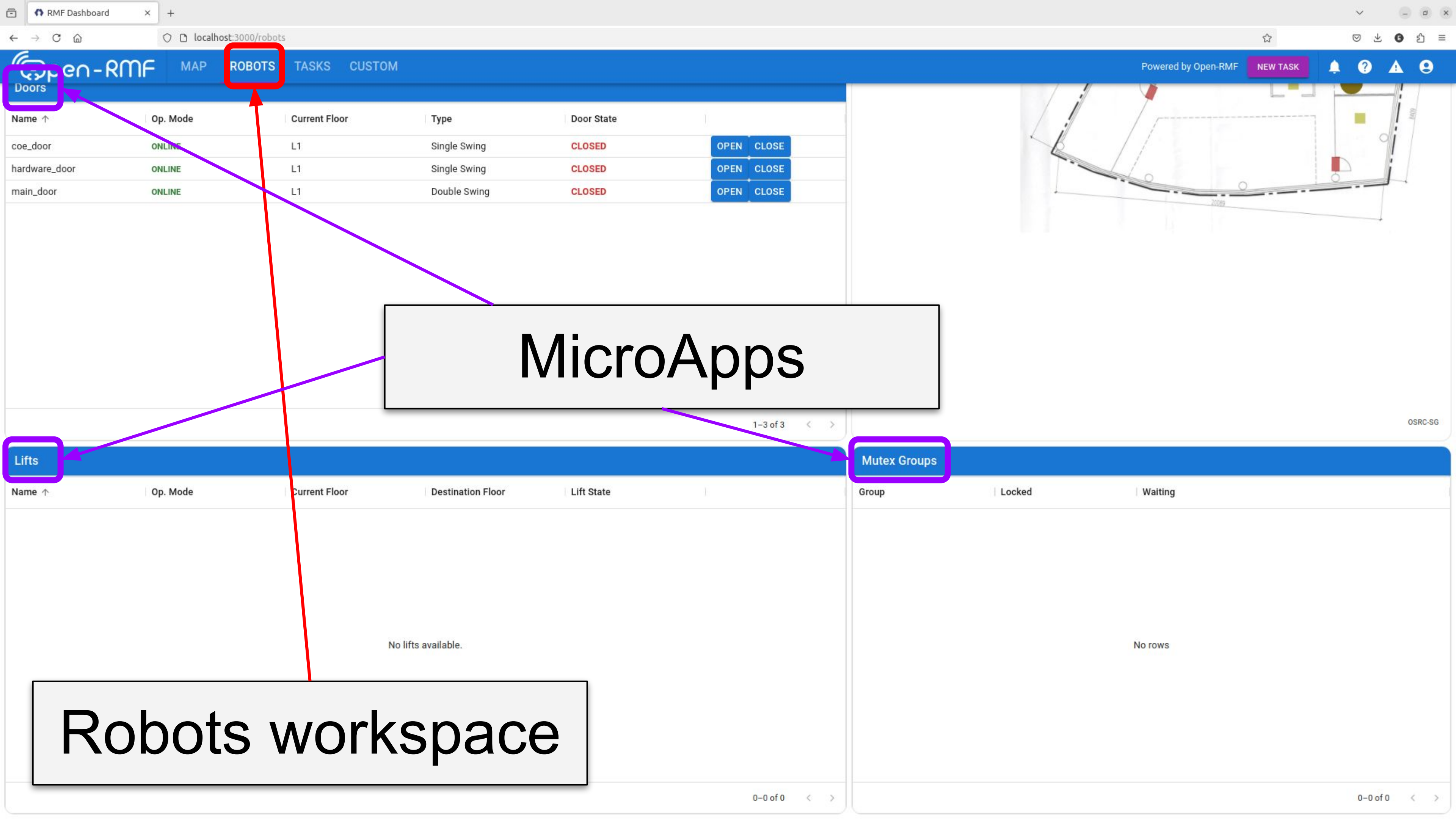
Workspaces

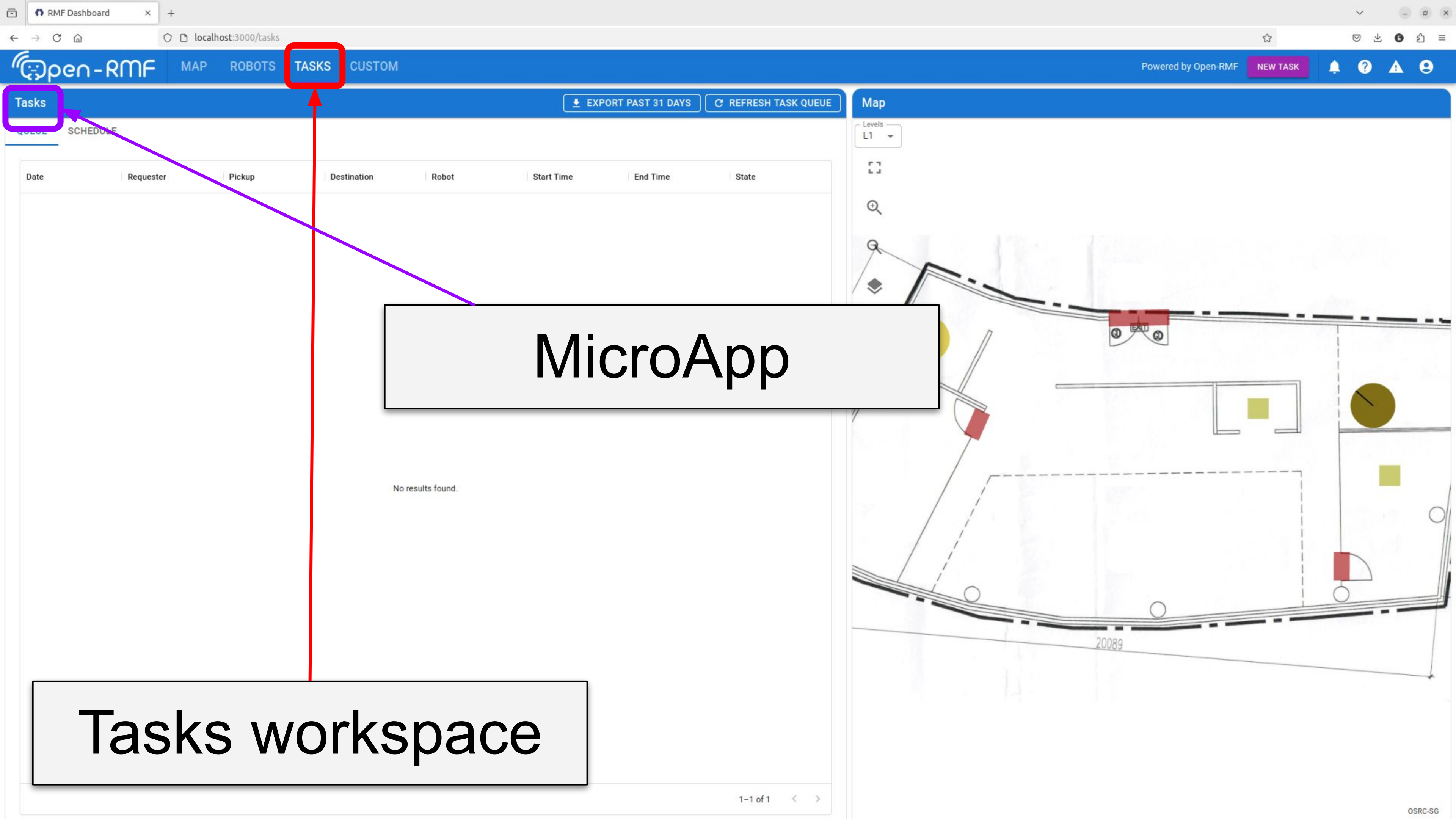


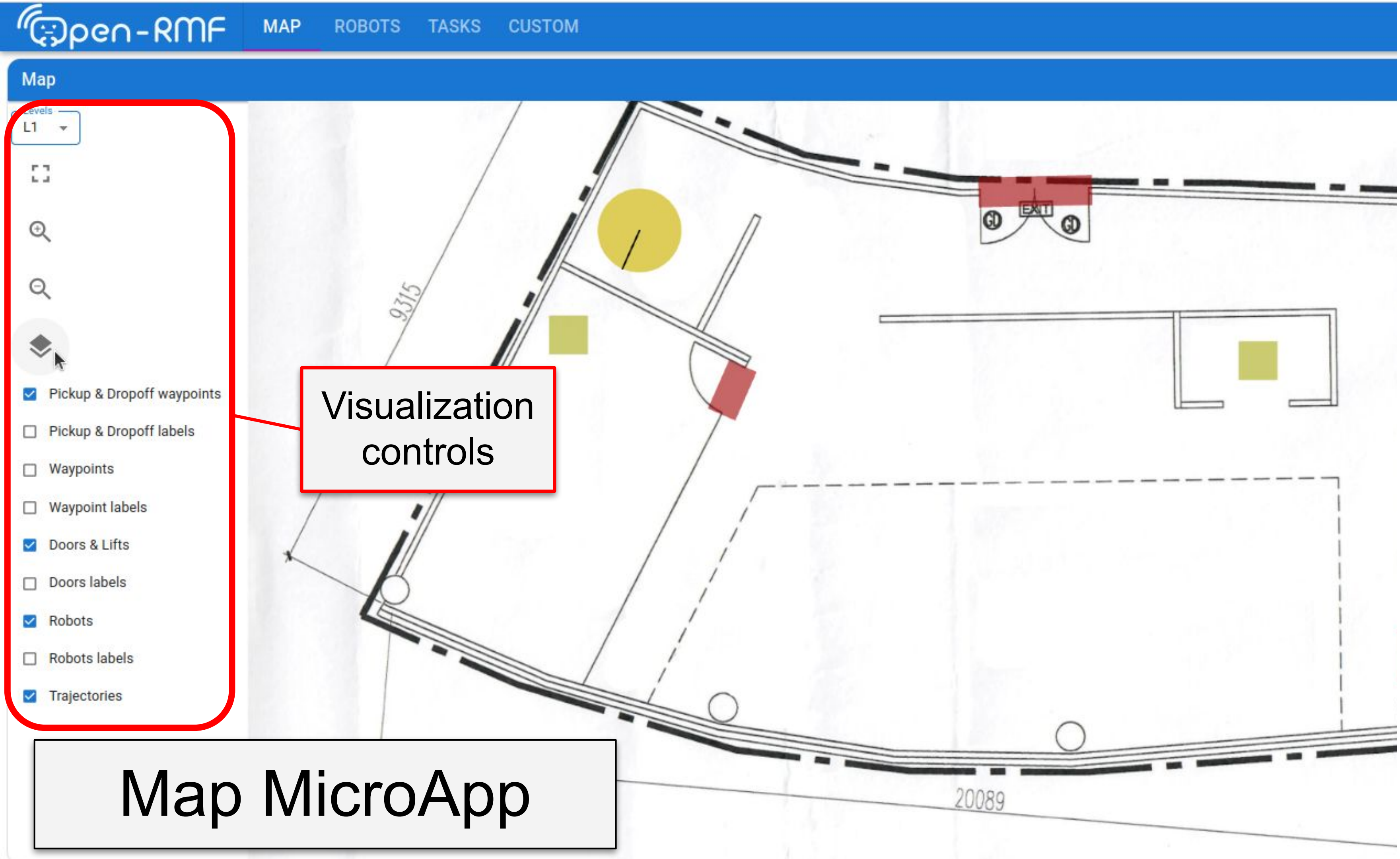
A Workspace is a React component that renders MicroApps based on a given layout, it makes it very easy to build a page by combining multiple MicroApps.

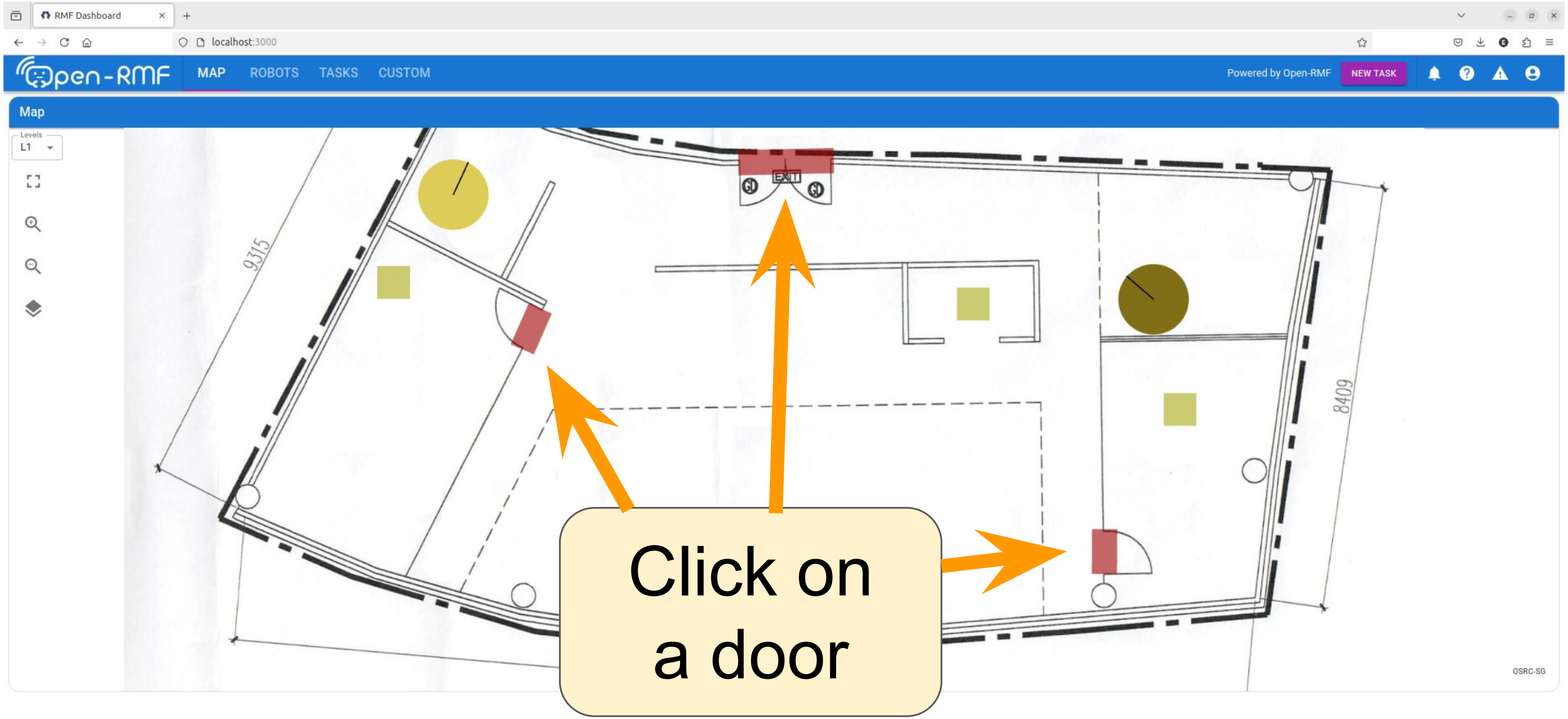

RMF Dashboard
localhost:3000
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Map
Levels
L1
9315
EXIT
8409
OSRC-SG
MicroApp
Map workspace


RMF Dashboard
localhost:3000/robots
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Robots
Name
Fleet
Est. Task Fin...
Level
Battery
Last Updated
Status
tinyRobot1 tinyRobot - L1 94.00% 1/1/1970, 1... IDLE
tinyRobot2 tinyRobot - L1 94.00% 1/1/1970, 1... CHARGING
1–2 of 2
Map
L1
MicroApps
Doors
Name
Op. Mode
Current Floor
Type
Door State
coe_door ONLINE L1 Single Swing CLOSED OPEN CLOSE
hardware_door ONLINE L1 Single Swing CLOSED OPEN CLOSE
main_door ONLINE L1 Double Swing CLOSED OPEN CLOSE
1–3 of 3
Robots workspace
20089
8409
OSRC-SG

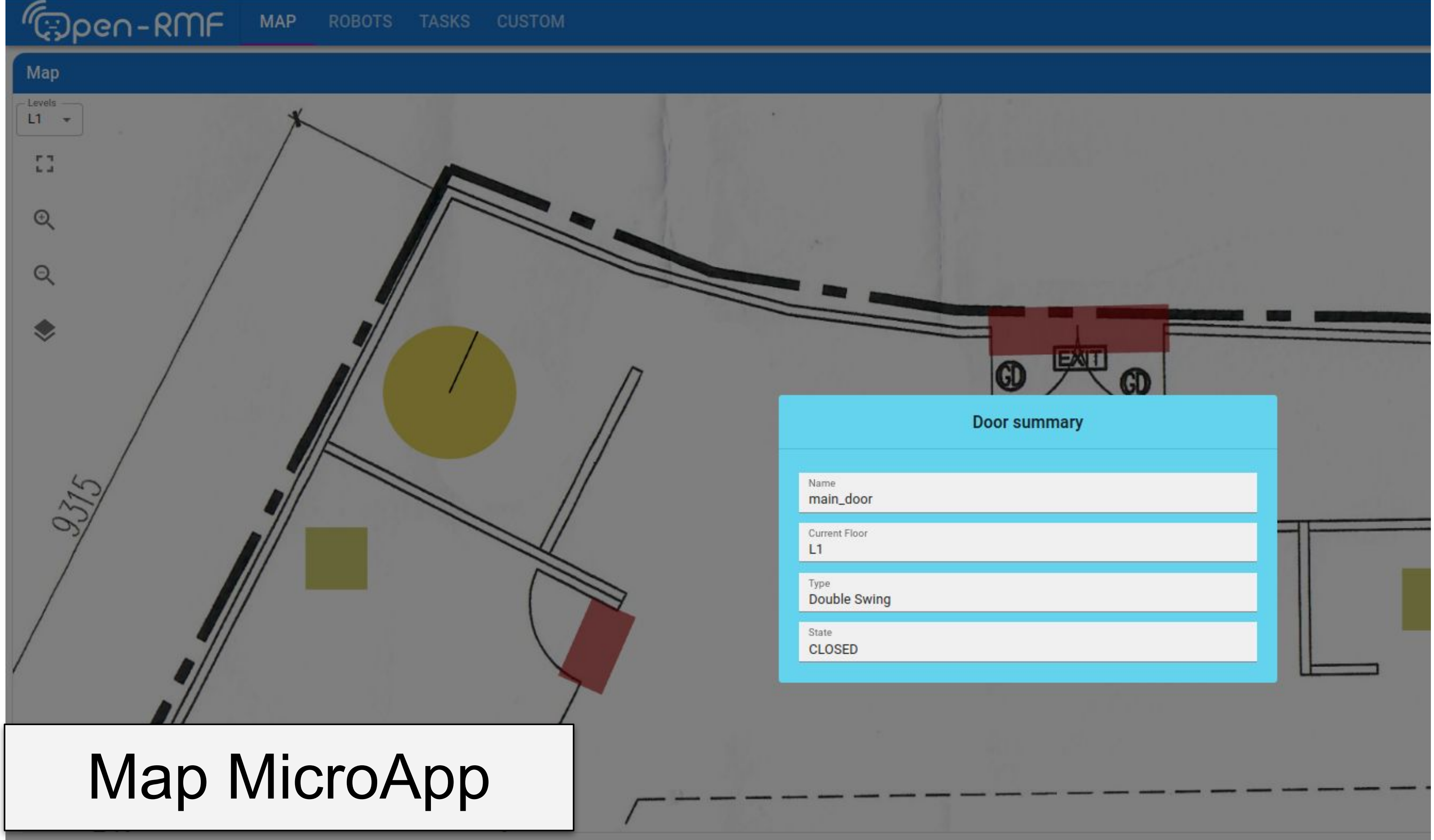

RMF Dashboard
localhost:3000/robots
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Doors
Name
Op. Mode
Current Floor
Type
Door State
coe_door
ONLINE
L1
Single Swing
CLOSED
OPEN
CLOSE
hardware_door
ONLINE
L1
Single Swing
CLOSED
OPEN
CLOSE
main_door
ONLINE
L1
Double Swing
CLOSED
OPEN
CLOSE
1–3 of 3
OSRC-SG
MicroApps
Lifts
Name
Op. Mode
Current Floor
Destination Floor
Lift State
No lifts available.
0–0 of 0
Mutex Groups
Group
Locked
Waiting
No rows
0–0 of 0
Robots workspace


RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
SCHEDULE
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
Map
Levels
L1
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
MicroApp
No results found.
Tasks workspace
1–1 of 1
20089
OSRC-SG

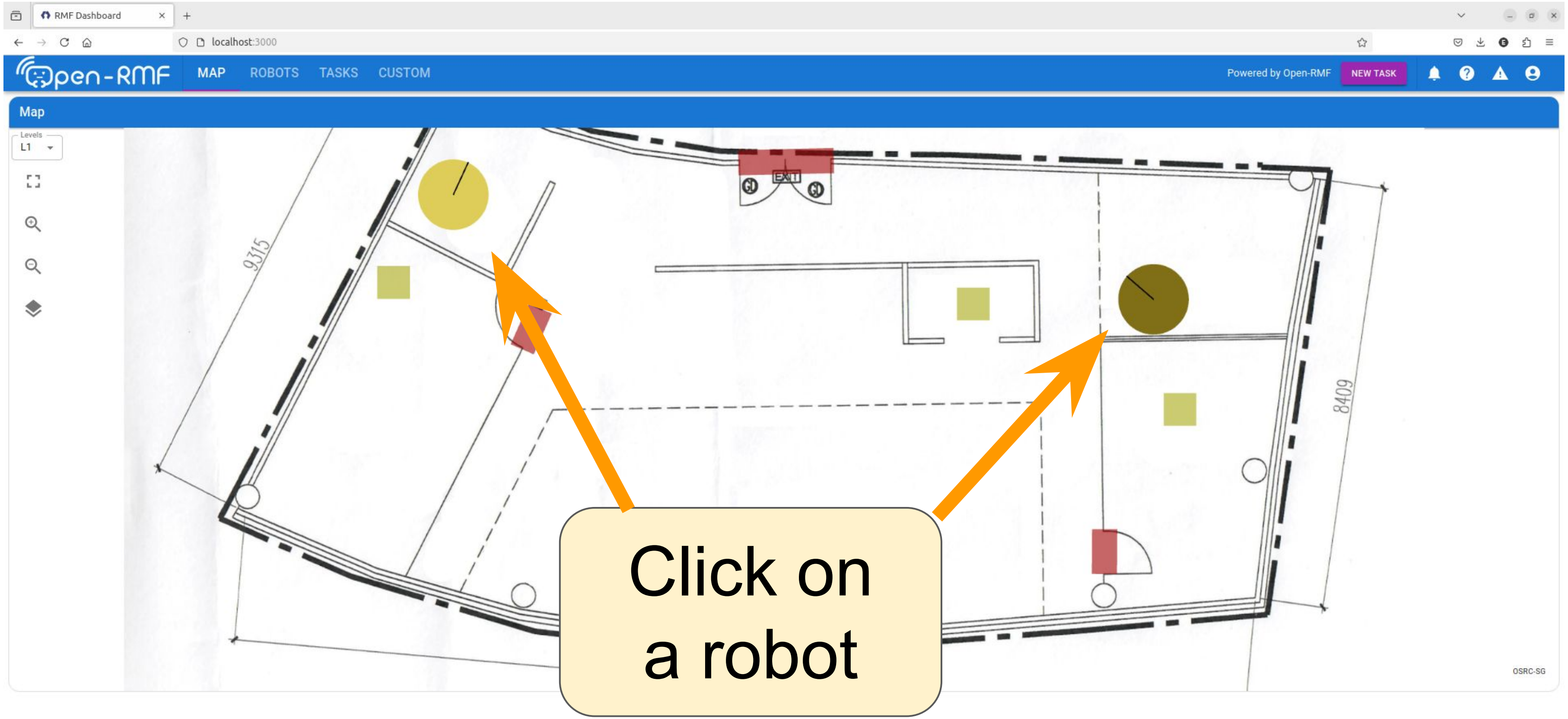

open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Map
Levels
L1
Pickup & Dropoff waypoints
Pickup & Dropoff labels
Waypoints
Waypoint labels
Doors & Lifts
Doors labels
Robots
Robots labels
Trajectories
Visualization controls
EXIT
9315
20089
Map MicroApp



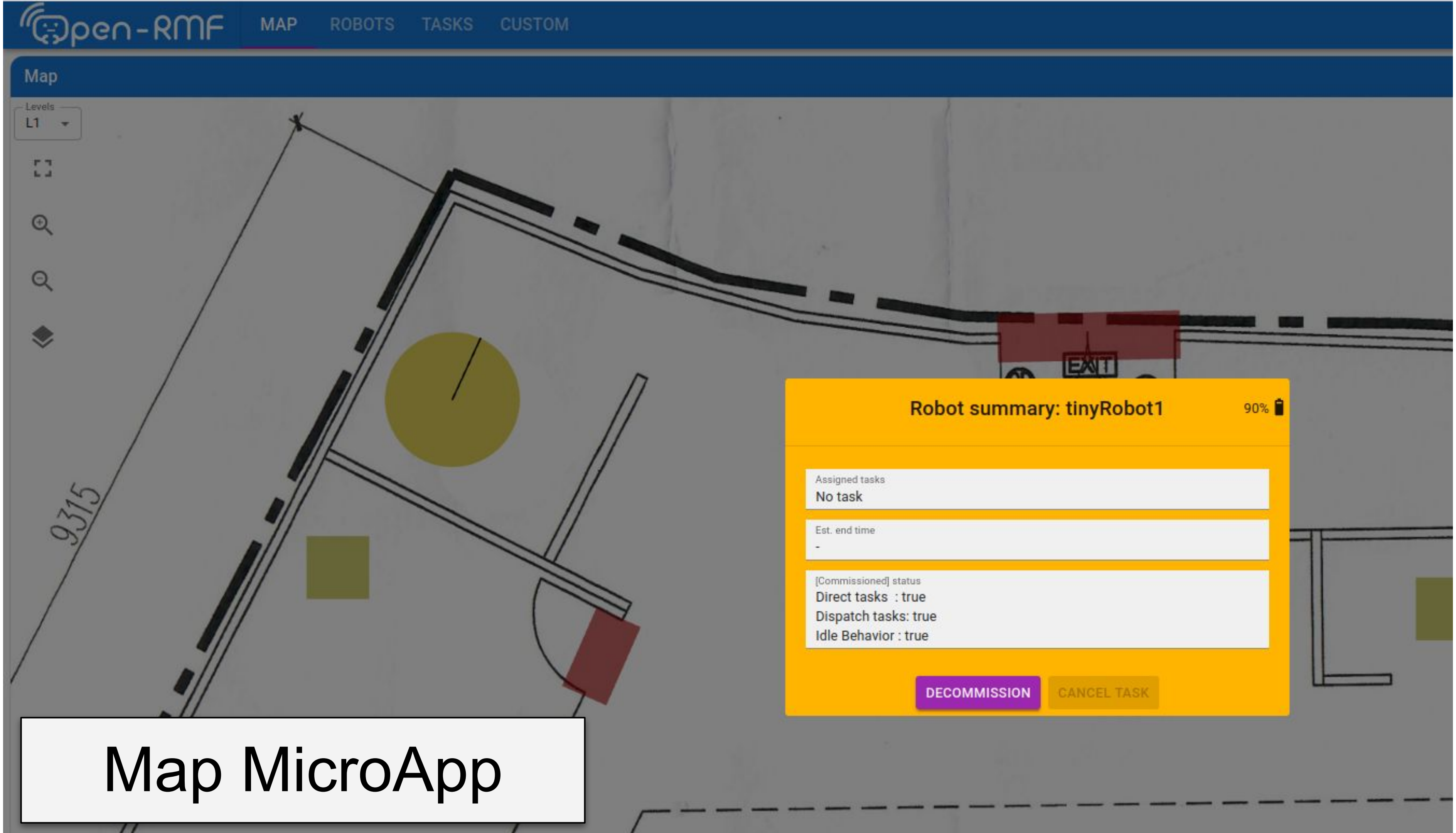
Map MicroApp



Map MicroApp



Map MicroApp



Map MicroApp

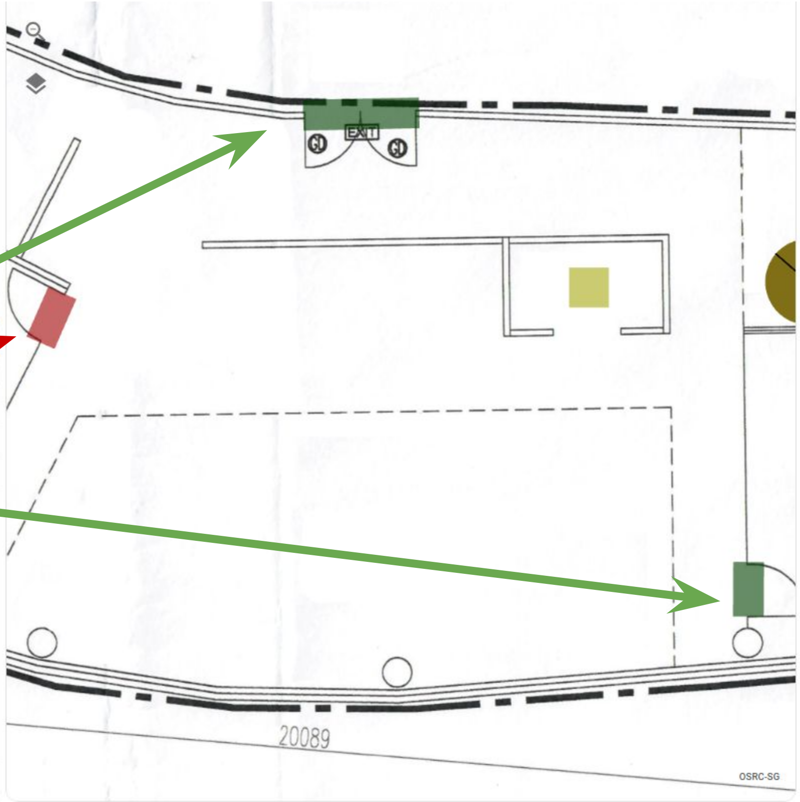
1–2 of 2

sorting / filtering

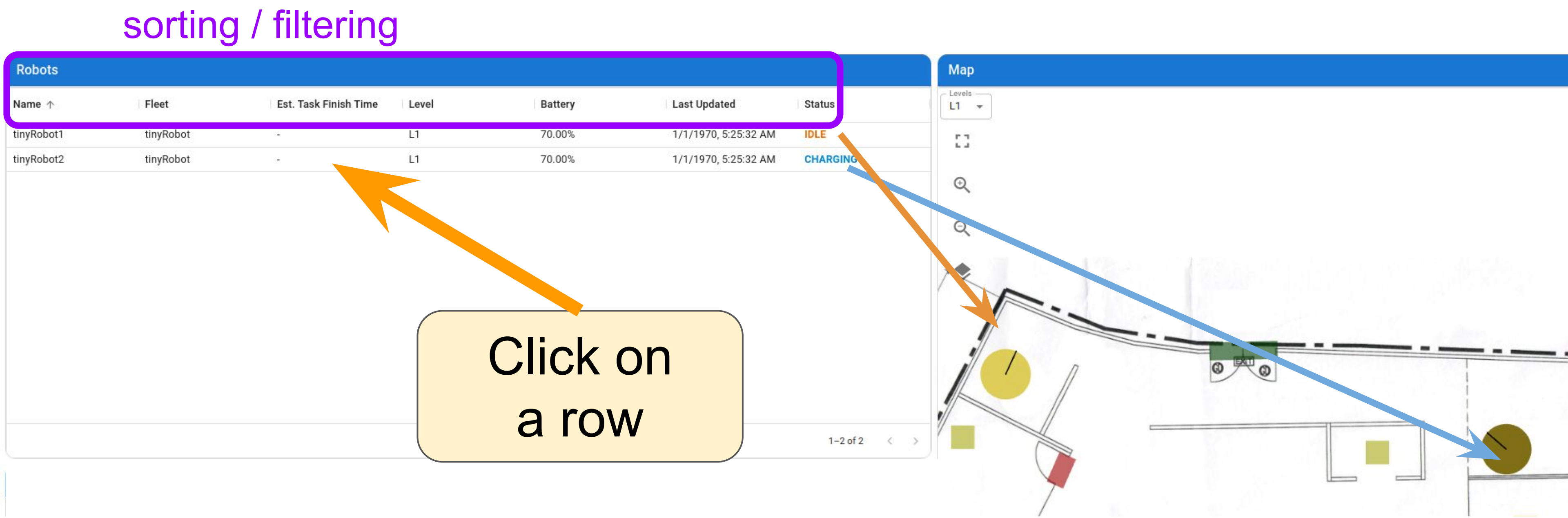
Doors

Name	Op. Mode	Current Floor	Type	Door State	
main_door	ONLINE	L1	Double Swing	OPEN	OPEN CLOSE
coe_door	ONLINE	L1	Single Swing	CLOSED	OPEN CLOSE
hardware_door	ONLINE	L1	Single Swing	OPEN	OPEN CLOSE

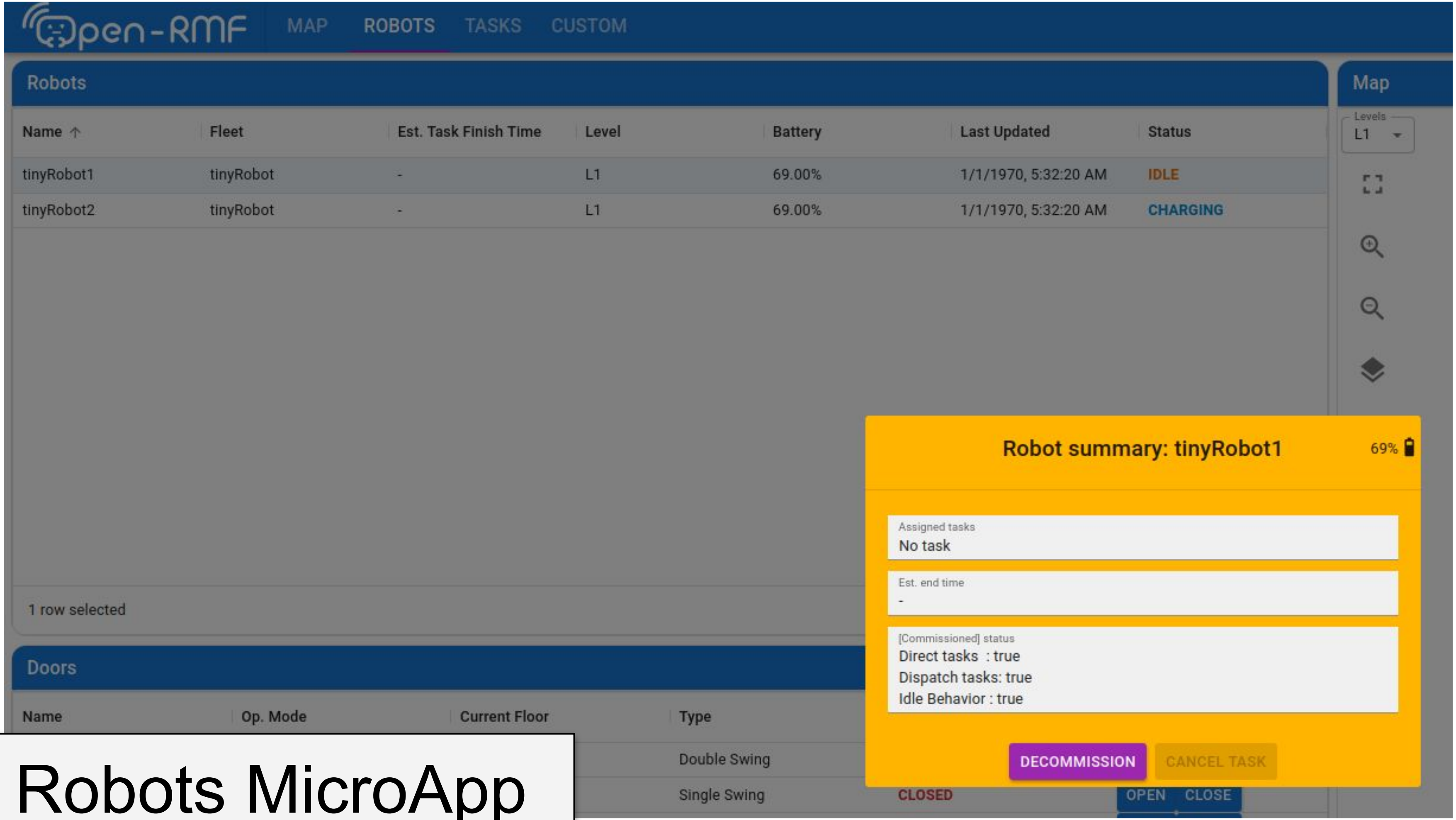
1–3 of 3



Doors MicroApp

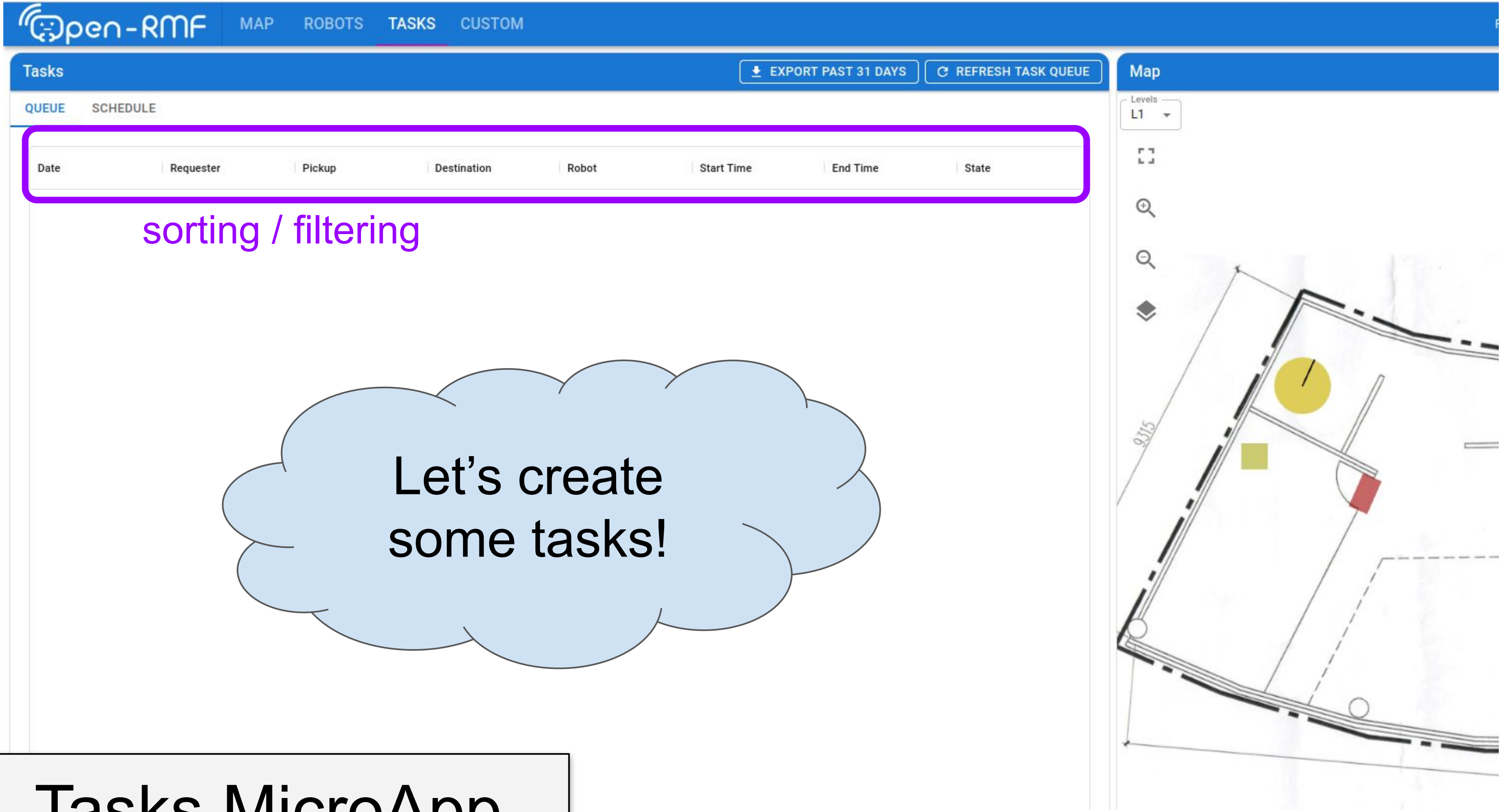


Robots MicroApp



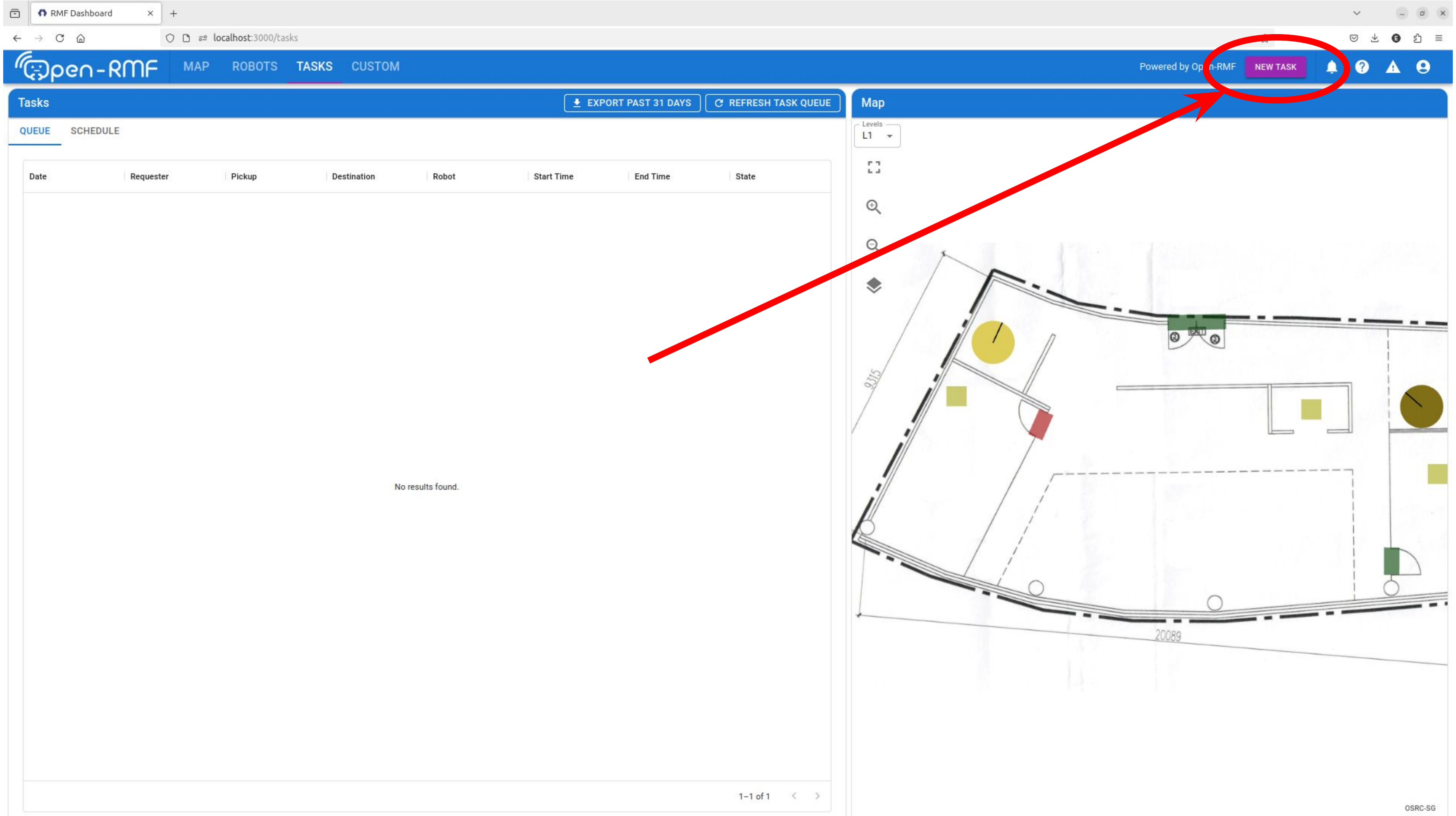
Robots MicroApp

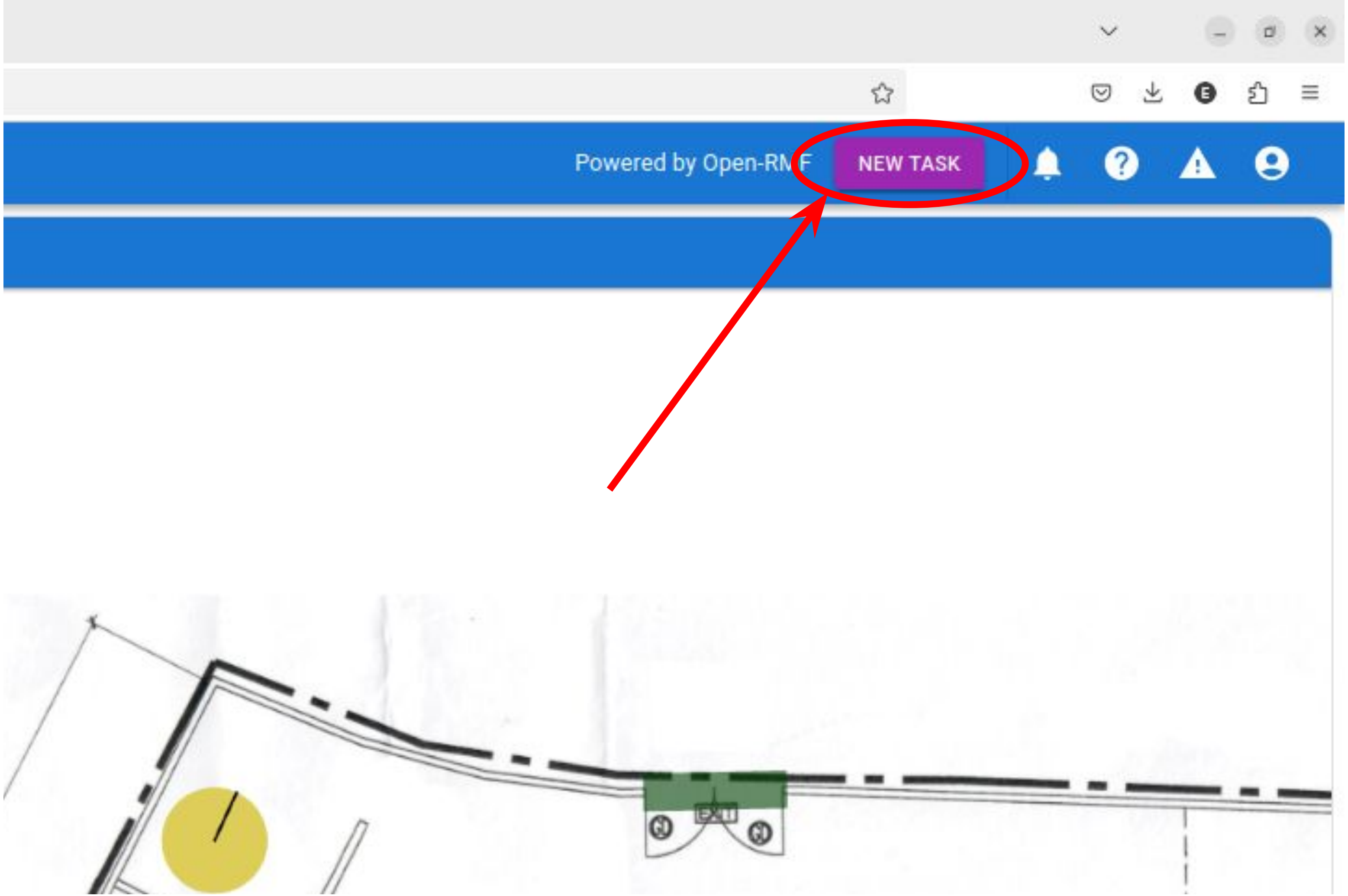
Tasks MicroApp



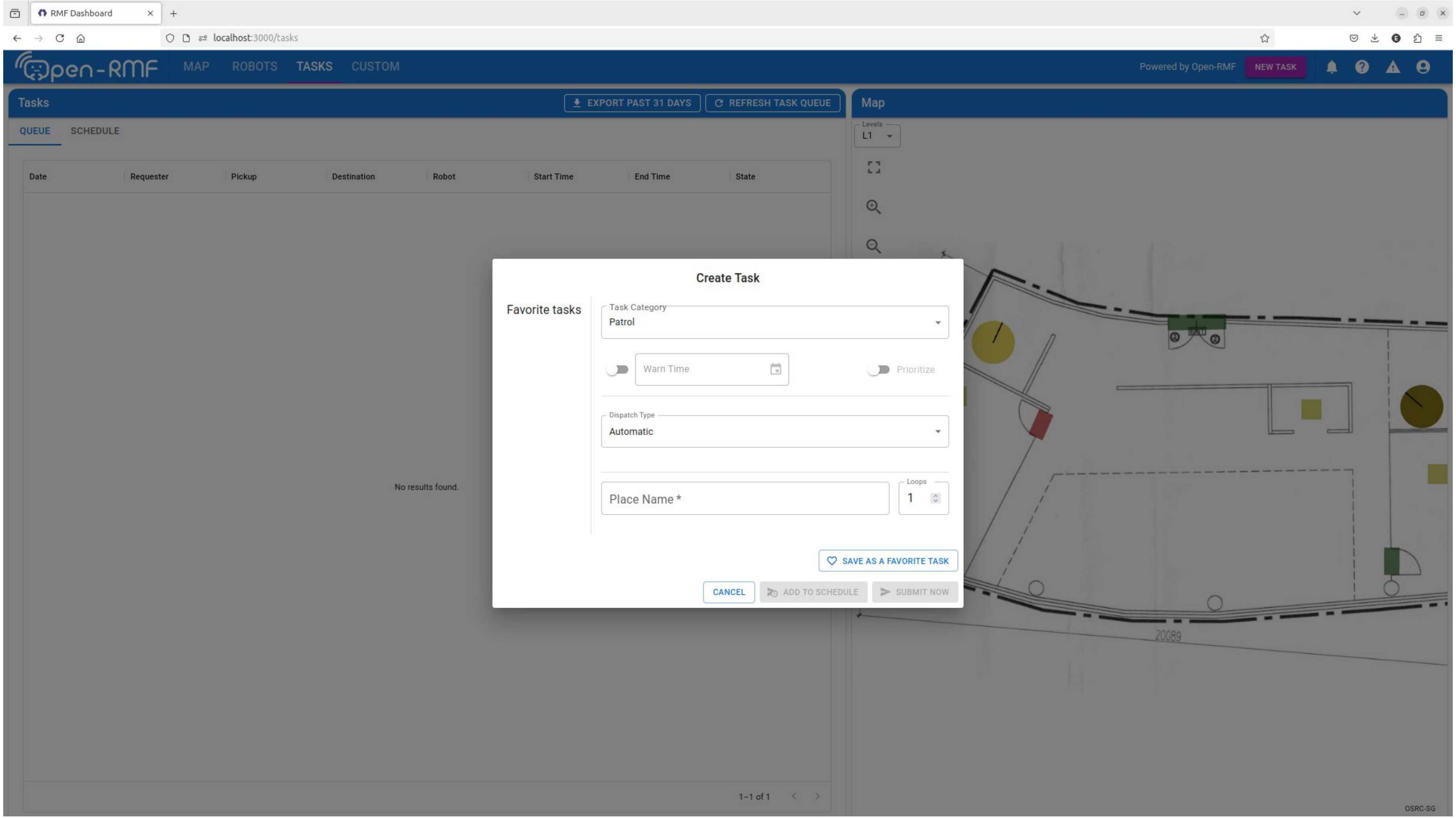
sorting / filtering

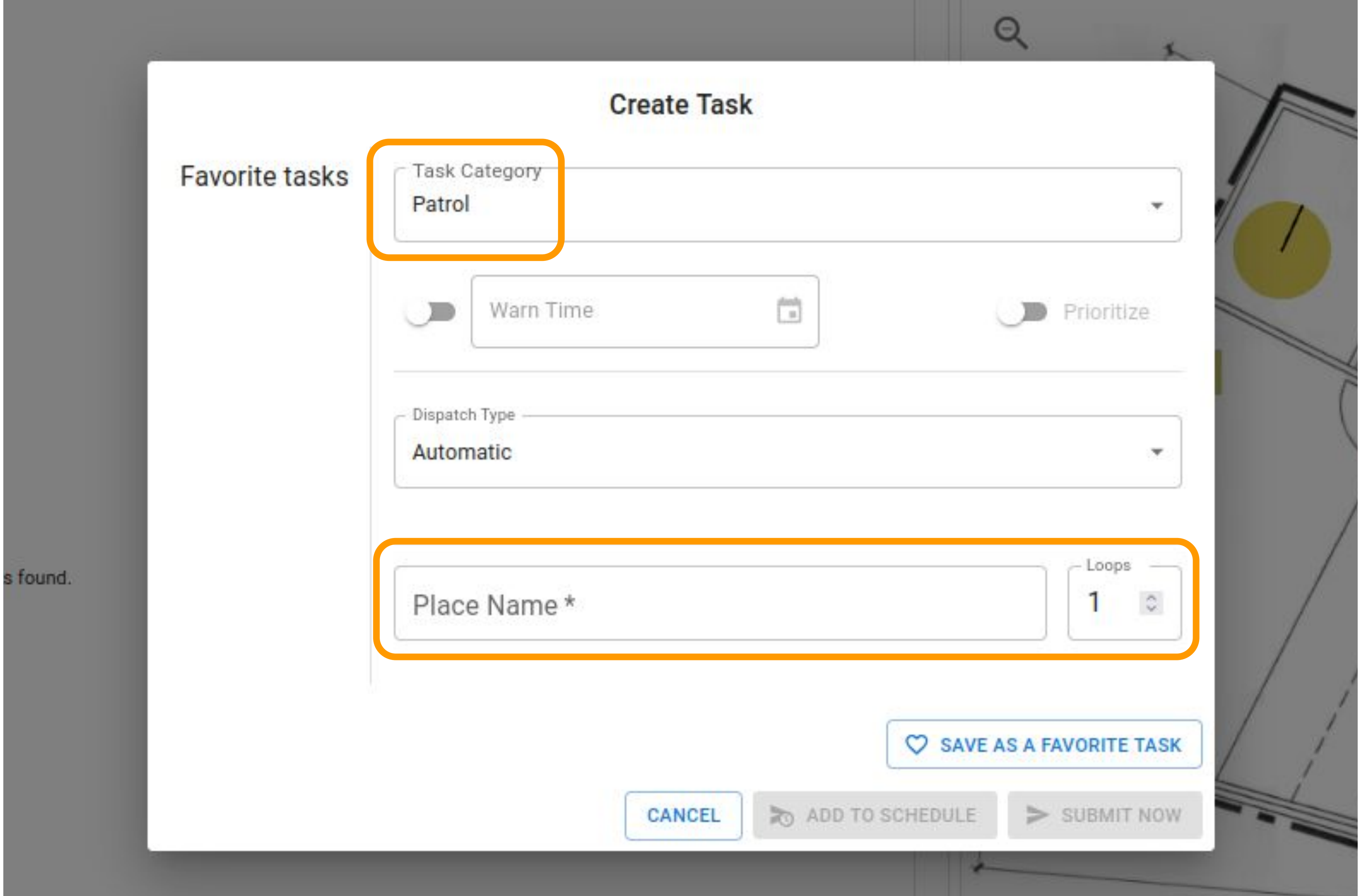
Let’s create some tasks!


RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
QUEUE
SCHEDULE
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
No results found.
1–1 of 1
Map
Levels
L1
9315
20089
OSRC-SG

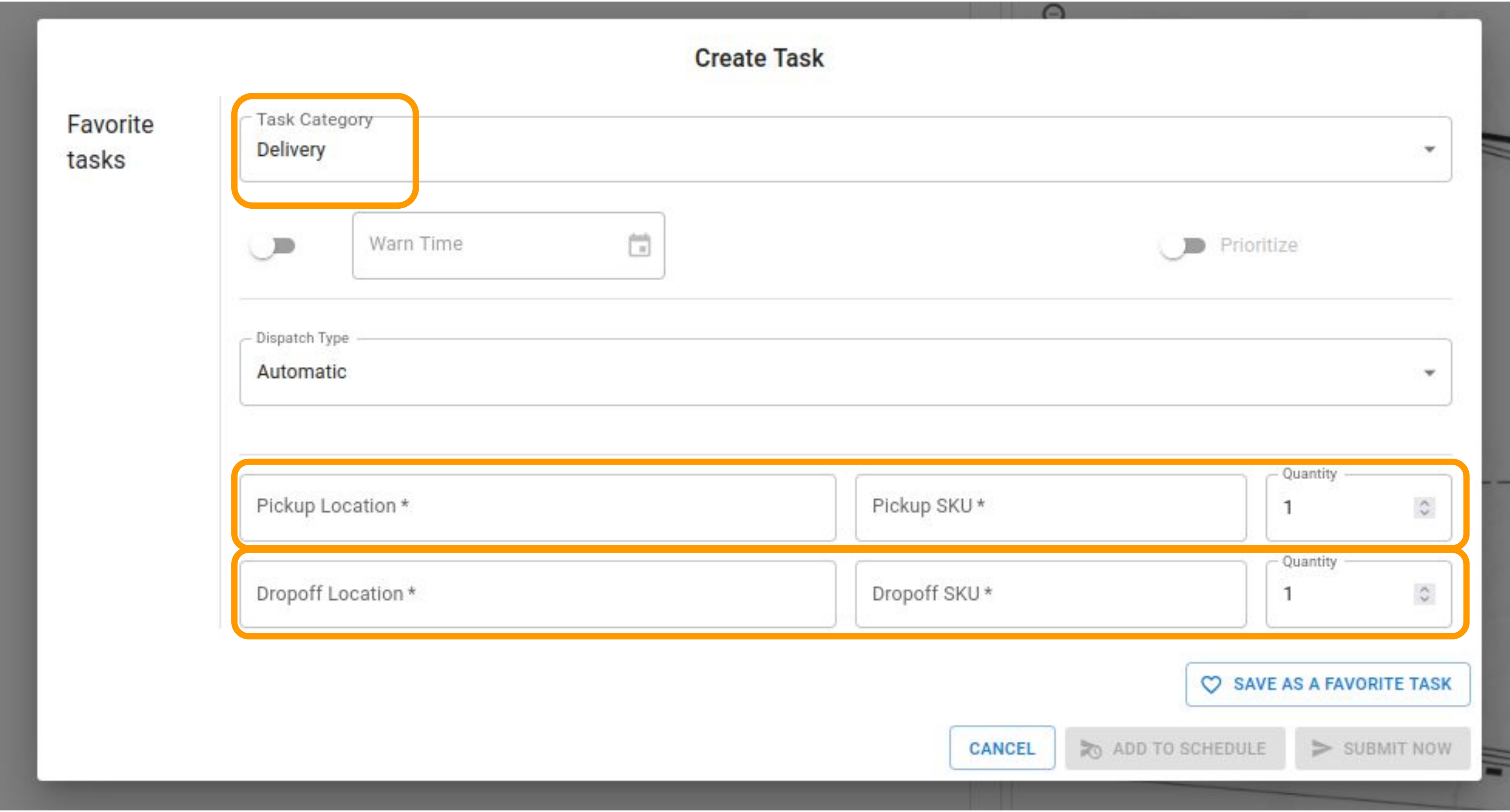


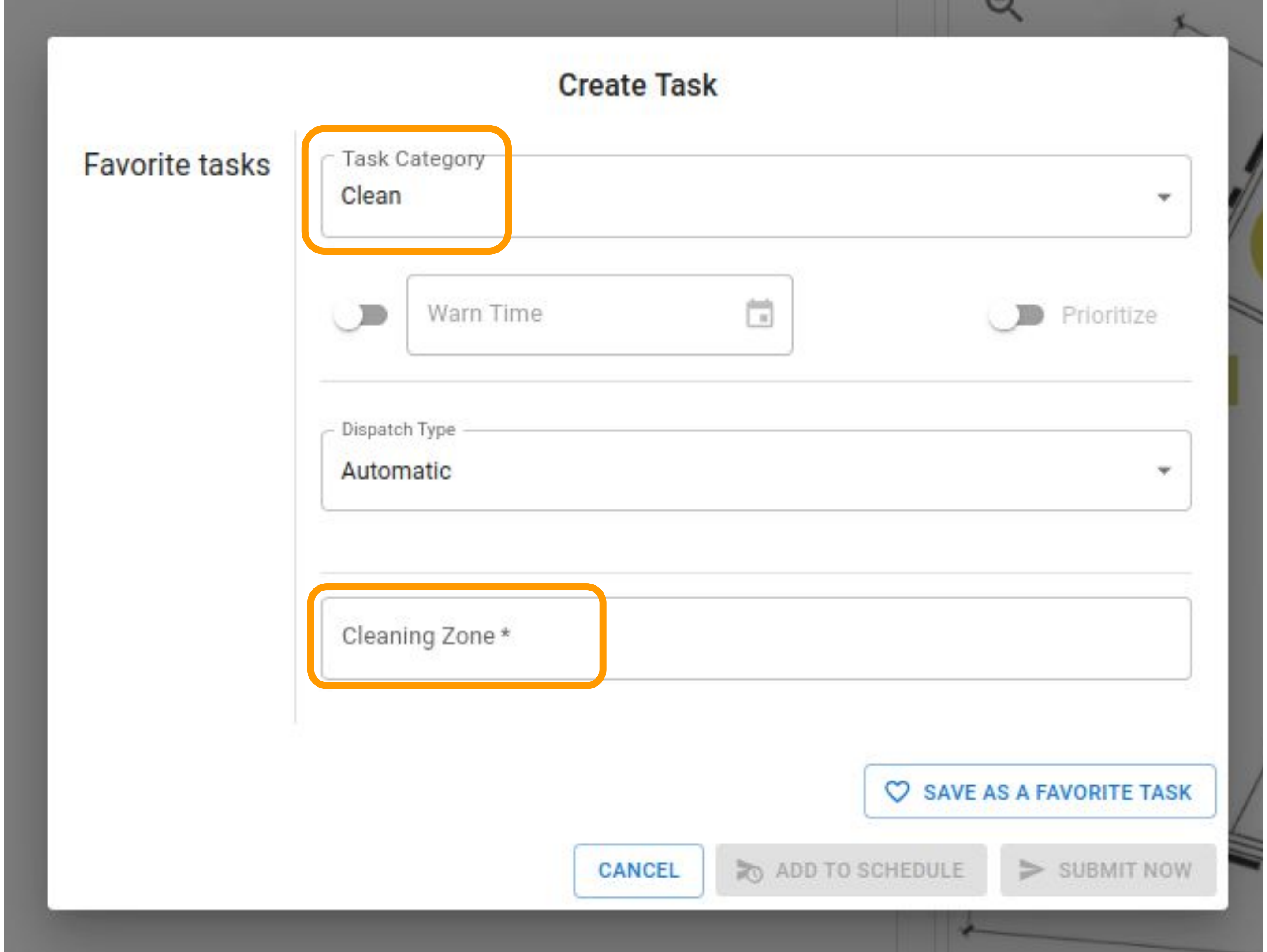
Powered by Open-RMF
NEW TASK


RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
QUEUE
SCHEDULE
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
No results found.
1–1 of 1
Map
Levels
L1
Create Task
Favorite tasks
Task Category
Patrol
Warn Time
Prioritize
Dispatch Type
Automatic
Place Name *
Loops
1
SAVE AS A FAVORITE TASK
CANCEL
ADD TO SCHEDULE
SUBMIT NOW
20089
OSRC-SG



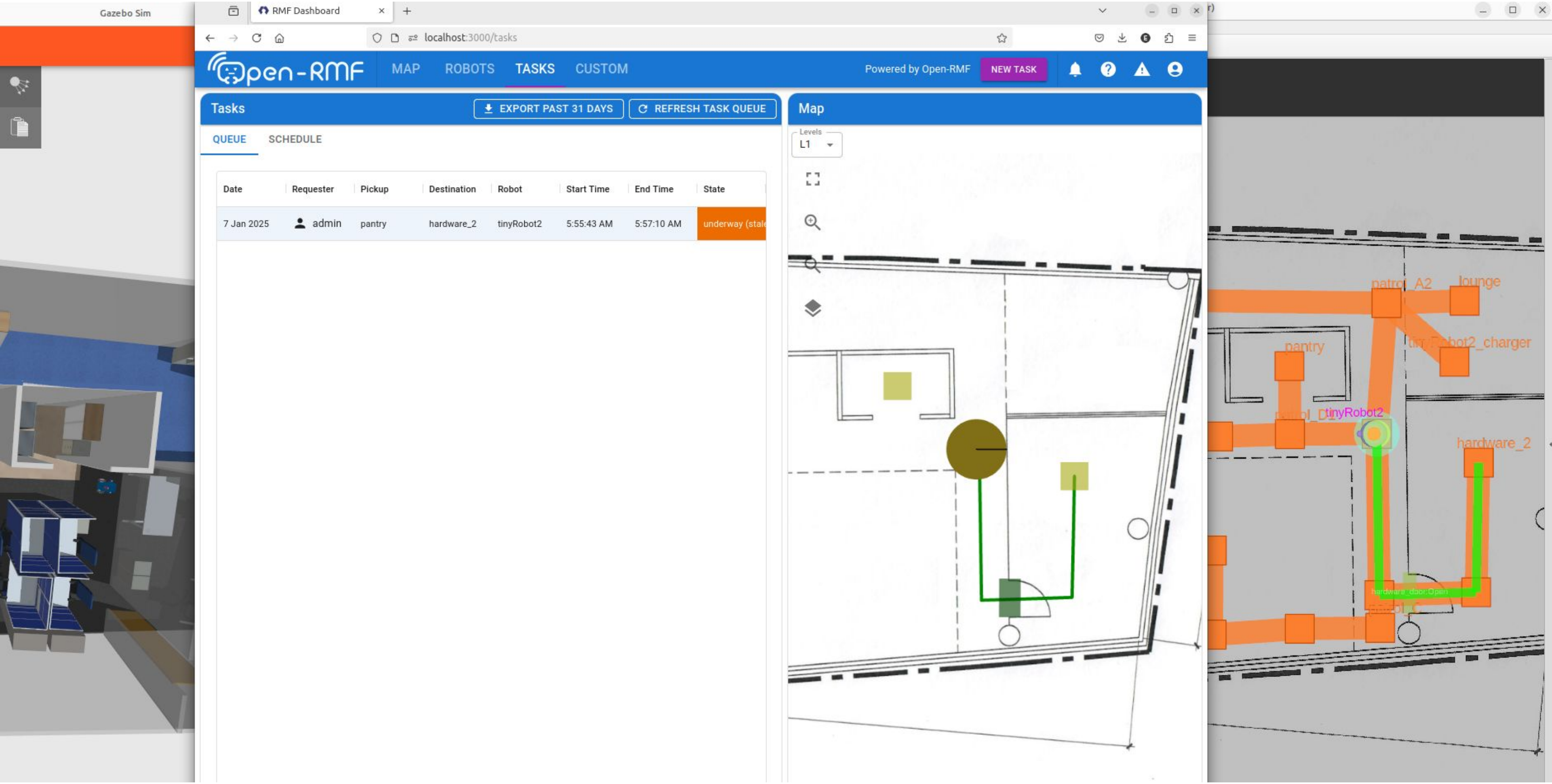
Create Task
Favorite tasks
Task Category
Patrol
Warn Time
Prioritize
Dispatch Type
Automatic
Place Name *
Loops
1
s found.
SAVE AS A FAVORITE TASK
CANCEL
ADD TO SCHEDULE
SUBMIT NOW


Create Task
Favorite tasks
Task Category
Delivery
Warn Time
Prioritize
Dispatch Type
Automatic
Pickup Location *
Pickup SKU *
Quantity
1
Dropoff Location *
Dropoff SKU *
Quantity
1
SAVE AS A FAVORITE TASK
CANCEL
ADD TO SCHEDULE
SUBMIT NOW


Create Task
Favorite tasks
Task Category
Clean
Warn Time
Prioritize
Dispatch Type
Automatic
Cleaning Zone *
SAVE AS A FAVORITE TASK
CANCEL
ADD TO SCHEDULE
SUBMIT NOW

Create the following tasks

Deliver a can of coke from **pantry** to **hardware_2**.

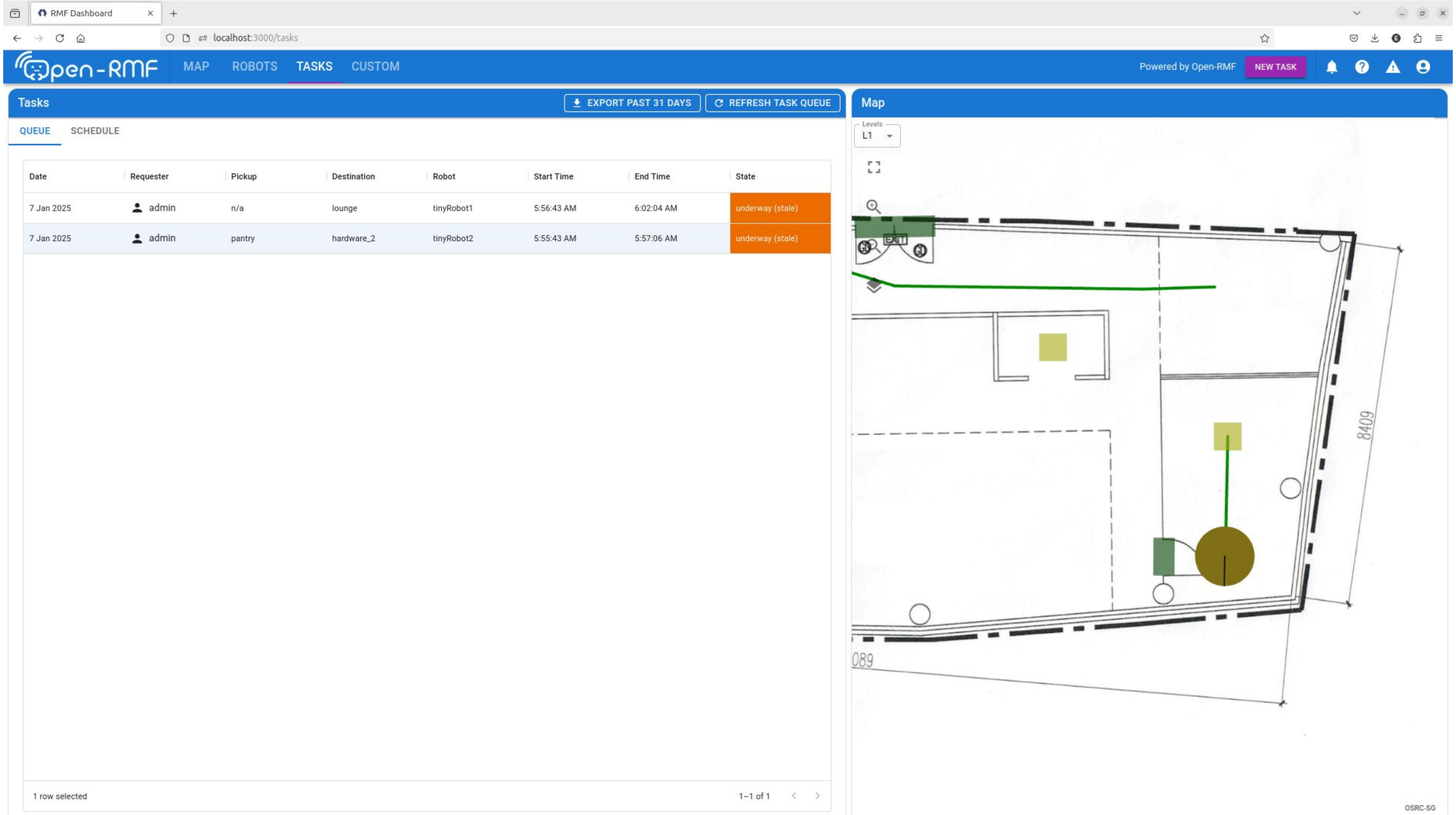


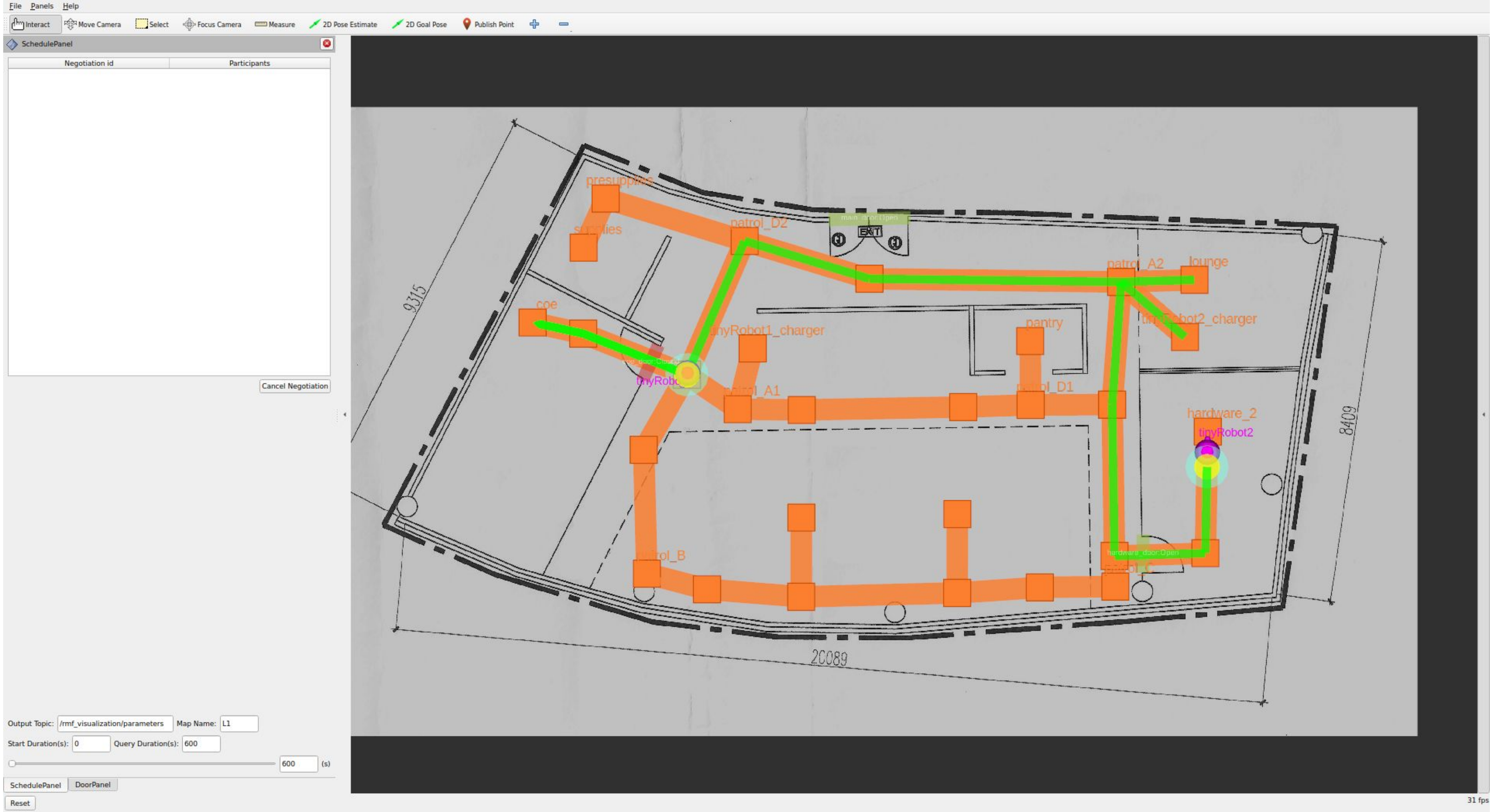
Gazebo Sim
RMF Dashboard
localhost:3000/tasks
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
QUEUE
SCHEDULE
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
7 Jan 2025
admin
pantry
hardware_2
tinyRobot2
5:55:43 AM
5:57:10 AM
underway (stale
Map
Levels
L1
patrol_A2
lounge
pantry
tinyRobot2
hardware_2

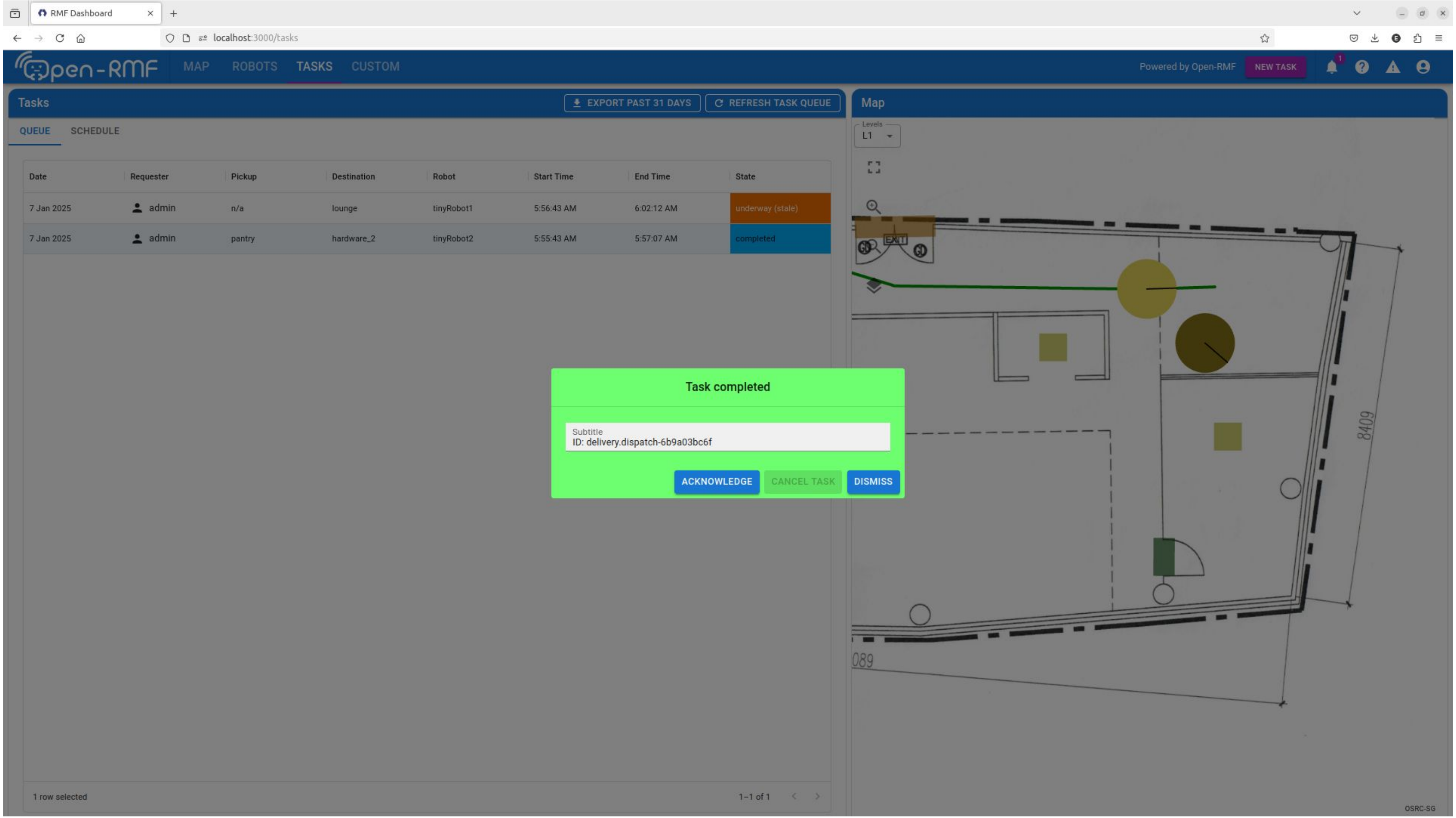
Create the following tasks

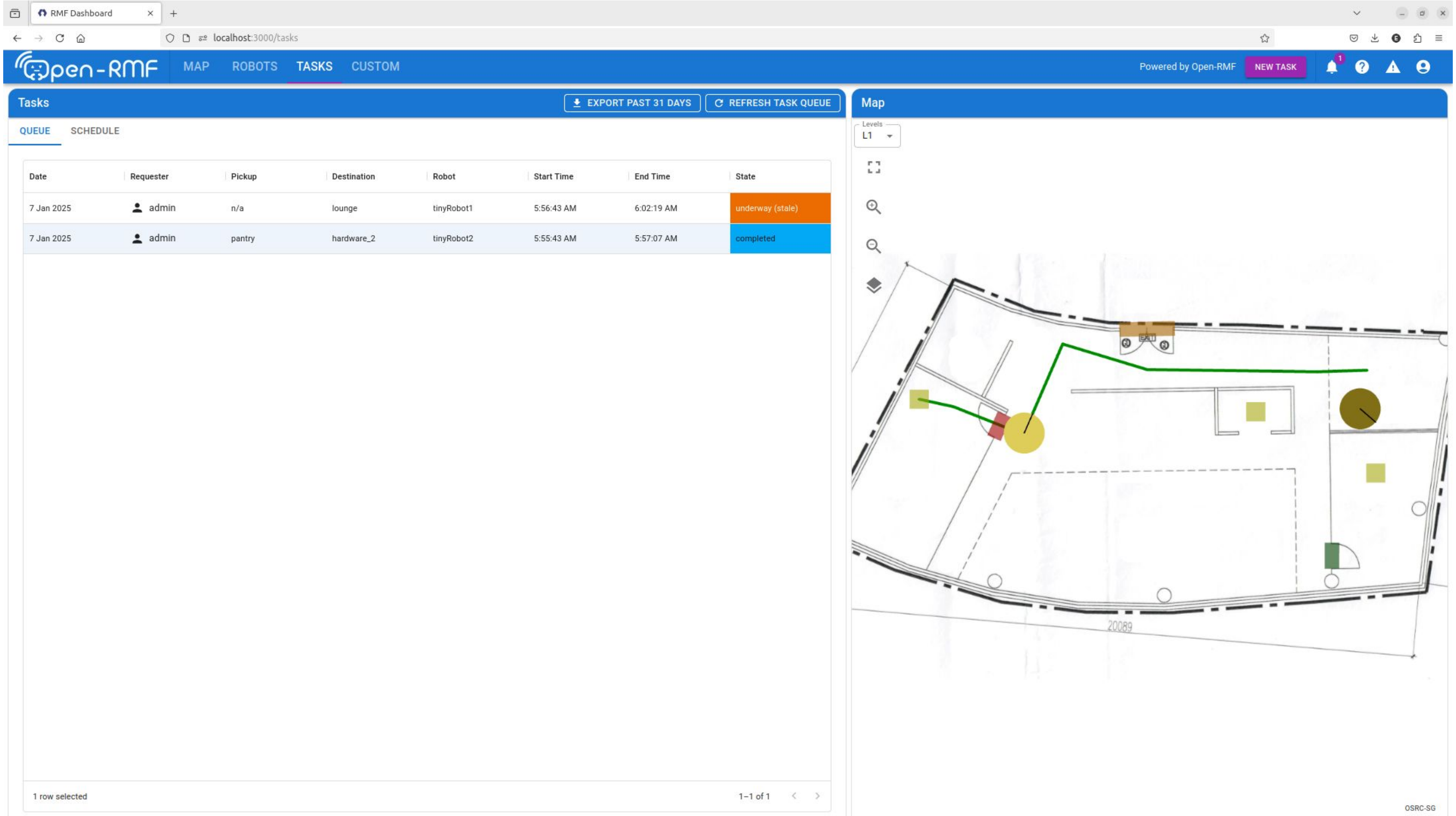
Deliver a can of coke from **pantry** to **hardware_2**.

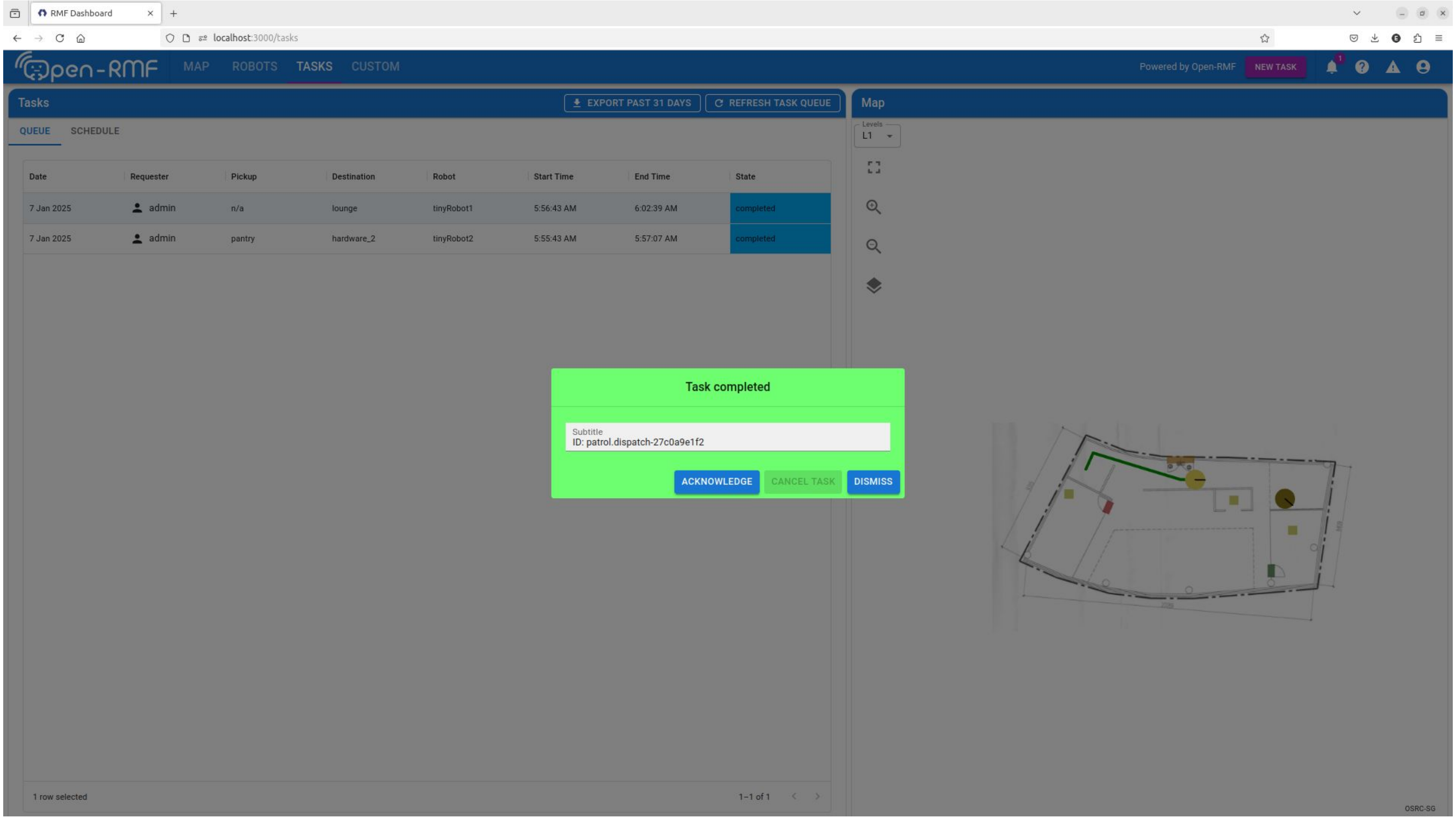
Move back and forth between **coe** and **lounge** for 3 times.


RMF Dashboard
localhost:3000/tasks
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
QUEUE
SCHEDULE
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
7 Jan 2025
admin
n/a
lounge
tinyRobot1
5:56:43 AM
6:02:04 AM
underway (stale)
7 Jan 2025
admin
pantry
hardware_2
tinyRobot2
5:55:43 AM
5:57:06 AM
underway (stale)
1 row selected
1–1 of 1
Map
Levels
L1
8409
089
OSRC-SG

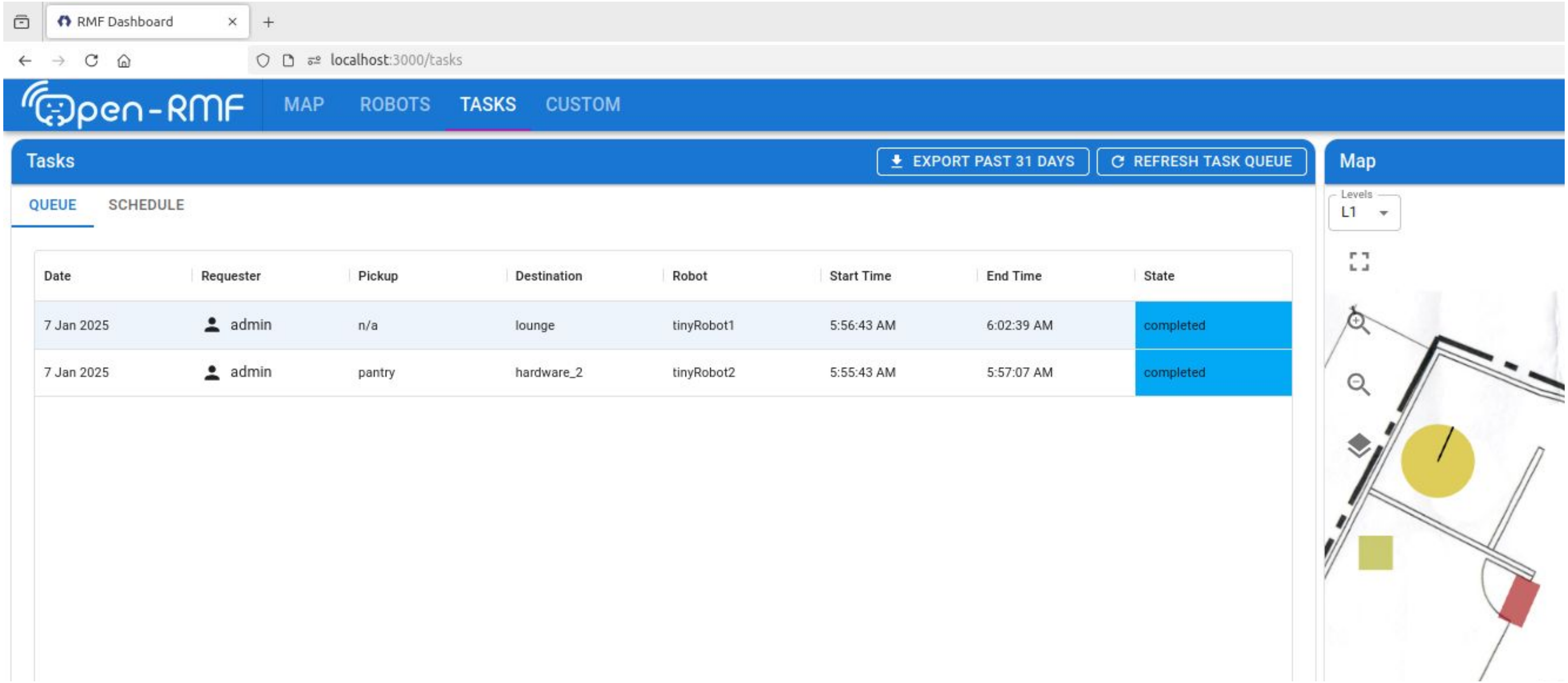

File
Panels
Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
SchedulePanel
Negotiation id
Participants
Cancel Negotiation
Output Topic:
/rmf_visualization/parameters
Map Name:
L1
Start Duration(s):
0
Query Duration(s):
600
600
(s)
SchedulePanel
DoorPanel
Reset
31 fps
presupplies
supplies
patrol_D2
coe
tinyRobot1_charger
patrol_A1
pantry
patrol_D1
patrol_A2
lounge
hardware_2
tinyRobot2
patrol_B
EXIT
9315
8409
20089

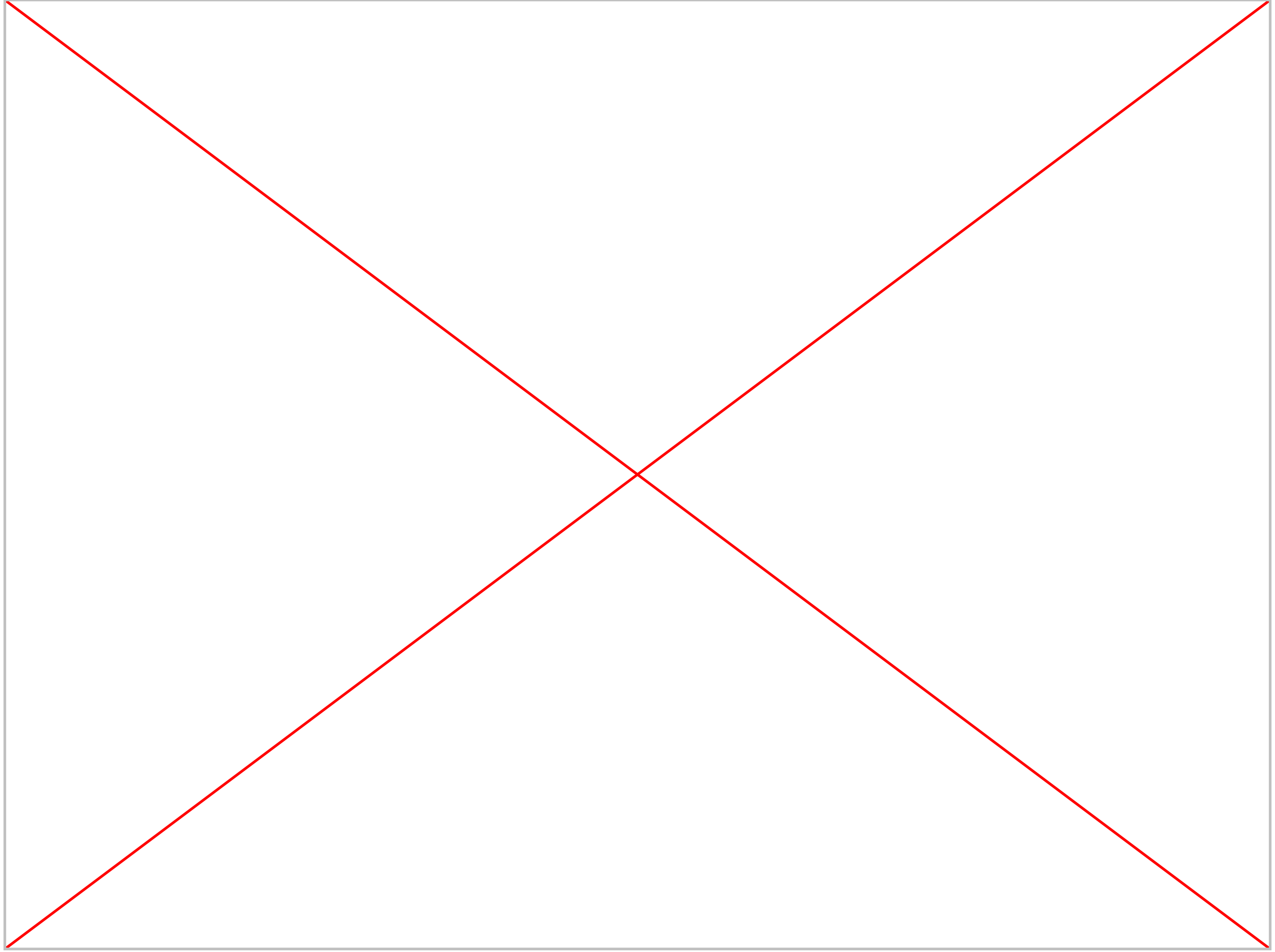

RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
QUEUE
SCHEDULE
Date | Requester | Pickup | Destination | Robot | Start Time | End Time | State
7 Jan 2025 | admin | n/a | lounge | tinyRobot1 | 5:56:43 AM | 6:02:12 AM | underway (stale)
7 Jan 2025 | admin | pantry | hardware_2 | tinyRobot2 | 5:55:43 AM | 5:57:07 AM | completed
1 row selected
1–1 of 1
Map
Levels
L1
Task completed
Subtitle
ID: delivery.dispatch-6b9a03bc6f
ACKNOWLEDGE
CANCEL TASK
DISMISS
8409
089
OSRC-SG

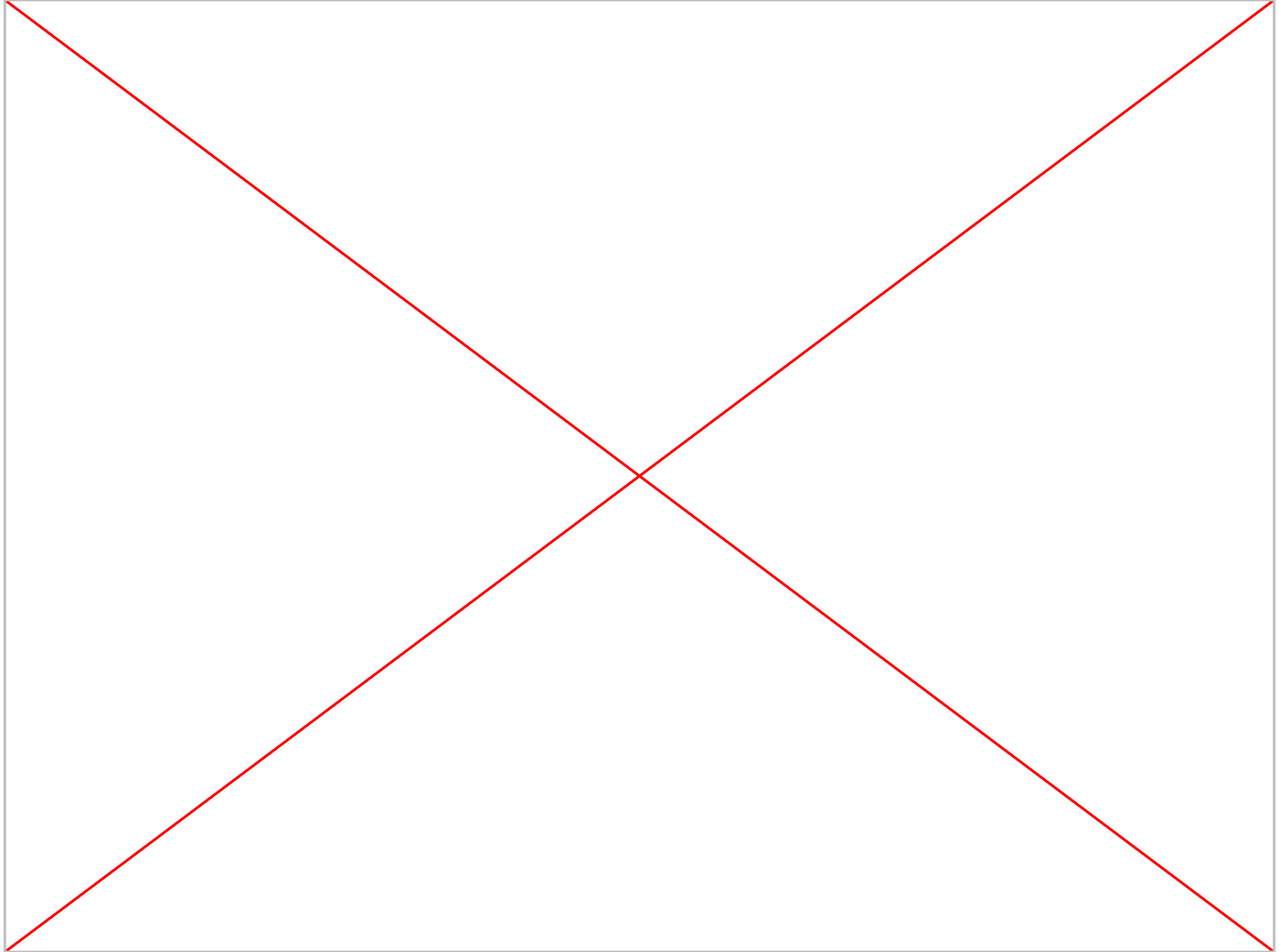

RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
QUEUE
SCHEDULE
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
7 Jan 2025
admin
n/a
lounge
tinyRobot1
5:56:43 AM
6:02:19 AM
underway (stale)
7 Jan 2025
admin
pantry
hardware_2
tinyRobot2
5:55:43 AM
5:57:07 AM
completed
1 row selected
1–1 of 1
Map
Levels
L1
20089
OSRC-SG



RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Powered by Open-RMF
NEW TASK
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
Map
QUEUE
SCHEDULE
Levels
L1
Date | Requester | Pickup | Destination | Robot | Start Time | End Time | State
7 Jan 2025 | admin | n/a | lounge | tinyRobot1 | 5:56:43 AM | 6:02:39 AM | completed
7 Jan 2025 | admin | pantry | hardware_2 | tinyRobot2 | 5:55:43 AM | 5:57:07 AM | completed
Task completed
Subtitle
ID: patrol.dispatch-27c0a9e1f2
ACKNOWLEDGE
CANCEL TASK
DISMISS
1 row selected
1–1 of 1
OSRC-SG


RMF Dashboard
localhost:3000/tasks
Open-RMF
MAP
ROBOTS
TASKS
CUSTOM
Tasks
EXPORT PAST 31 DAYS
REFRESH TASK QUEUE
Map
QUEUE
SCHEDULE
Levels
L1
Date
Requester
Pickup
Destination
Robot
Start Time
End Time
State
7 Jan 2025
admin
n/a
lounge
tinyRobot1
5:56:43 AM
6:02:39 AM
completed
7 Jan 2025
admin
pantry
hardware_2
tinyRobot2
5:55:43 AM
5:57:07 AM
completed





Custom Compose Tasks

Users would sometimes like to send out specific and identical tasks throughout the deployment.

Instead of clicking through task creation for all the different parameters each time, task json files can be uploaded and sent.

Here is an example.

```
{
  "category": "compose",
  "phases": [
    {"activity":
      {"category": "go_to_place",
       "description": "coe"
      }
    },
    {"activity":
      {"category": "go_to_place",
       "description": "pantry"
      }
    }
  ]
}
```

Exercise 1

Simulate the hotel world

Dispatch two tasks with Open RMF web:

- Patrol once from “restaurant” to “L3_master_suite”
- Clean the zone “clean_lobby”

Record a screenshot of RMF Dashboard, RViz and Gazebo.

Look for **interesting events** i.e. situations with two robots sharing the same space (corridor, door, lift).

Exercise 2

Simulate the airport terminal world

Dispatch three tasks with Open RMF web (patrol, delivery, clean - see the information for each task in the demonstration slides or create your own tasks).

Record a screenshot of RMF Dashboard, RViz and Gazebo.

Look for **interesting events** i.e. situations with two robots sharing the same space (corridor, door, lift).

Exercise 3

Simulate the clinic world

Dispatch two patrol tasks with Open RMF web (see the information for each task in the demonstration slides or create your own tasks).

Record a screenshot of RMF Dashboard, RViz and Gazebo.

Look for **interesting events** i.e. situations with two robots sharing the same space (corridor, door, lift).

Qualification of screenshot videos

They should feature some **interesting events** i.e. situations with two robots sharing the same space (corridor, door, lift).

For the visualization in the video:

- **Minimum**: RMF web browser window
- **Fair**: RMF web + RViz + Gazebo windows
- **Good**: moving viewpoint in the windows for better showing the robot tasks
- **Excellent**: adding subtitles for description of situations (e.g. traffic conflicts)

References

- Open-RMF web